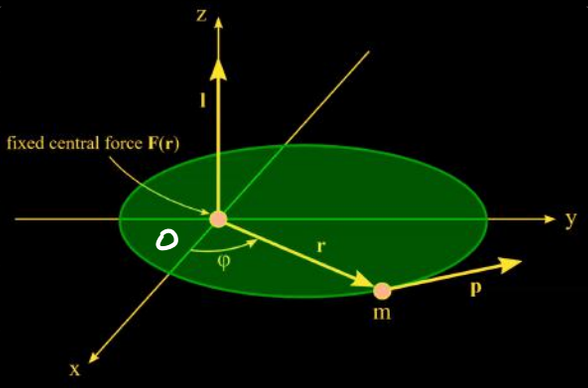
Angular Momentum Conservation In Central Forces:



Recall that a central force is of the form $\vec{F} = F(r)\hat{r}$. Earlier we saw that under action of a central force alone, mechanical energy of a particle remains conserved because all central forces are conservative forces. Now we are going to prove that under action of all such forces, angular momentum about the centre of the field always remains conserved. This can be easily seen as follows: Let O be the centre of a central field. Now recall that for any arbitrary particle torque about any arbitrary point O is related to its angular momentum as:

$$\vec{\tau}_o = \frac{d\vec{L}_o}{dt}, \quad \text{Also } \vec{\tau}_o = \vec{r} \times \vec{F}$$

however since it is a central force, $\vec{r}$ is collinear with $\vec{F}$ therefore $\vec{r} \times \vec{F} = 0$

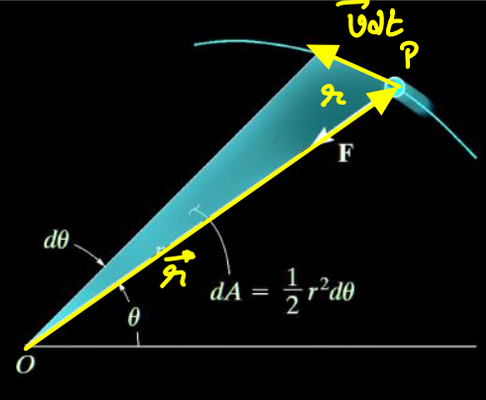
Thus:

$$\vec{\tau}_o = 0 = \frac{d\vec{L}_o}{dt} \Rightarrow \vec{L}_o \text{ is constant.}$$

Planar Nature Of Orbits In Central Fields:

Angular momentum of any particle wrt centre of the central field is given as $\vec{L} = \vec{r} \times m\vec{v}$ which is perpendicular to the plane containing $\vec{r}$ and $\vec{v}$. Now since angular momentum is conserved, it means the direction of the normal to the plane containing $\vec{r}$ and $\vec{v}$ is also fixed. Thus any particle moving under action of central forces alone must follow a planar trajectory.

Areal Velocity In Central Force Motion:



The area swept by line joining centre of the field to the particle per unit time is called areal velocity, i.e if the line sweeps out an area dA in time dt, areal velocity is simply dA/dt. We will now prove that if

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A particle is moving in a central field, its areal velocity is constant and related to its angular momentum as $dA/dt = |\vec{L}|/2m$.

Recall that area of any triangle is $\frac{1}{2}$ times | cross product of any two sides |.

Now let a particle in central force motion be located at a position vector $\vec{r}$ while moving with a velocity $\vec{v}$, then in a time dt, Area swept by it is given as: $d\vec{A} = \frac{1}{2}\vec{r} \times \vec{v}\,dt$, here we define $d\vec{A}$ as a vector whose magnitude is equal to the swept area and direction is along angular momentum (i.e $\vec{r} \times \vec{v}$).

Thus we can also write:

$$m\,d\vec{A} = \frac{1}{2}\vec{r} \times m\vec{v}\,dt \quad (1)$$

$$\Rightarrow \frac{d\vec{A}}{dt} = \frac{1}{2}\frac{\vec{r} \times \vec{p}}{m} = \frac{\vec{L}}{2m} \qquad (\vec{p}: \text{linear momentum i.e } m\vec{v}).$$

or $\boxed{\frac{d\vec{A}}{dt} = \frac{\vec{L}}{2m}}$ or $\boxed{\frac{dA}{dt} = \frac{|\vec{L}|}{2m}}$,

Now since angular momentum $\vec{L}$ is constant, areal velocity must also be constant i.e line joining the particle to the centre of the field sweeps out equal areas in equal intervals of time.

→ $\frac{dA}{dt}$ can also be written as $\frac{1}{2}r^2\frac{d\theta}{dt}$ because, $\frac{dA}{dt} = \frac{1}{2}|\vec{r} \times \vec{v}|$

$$= \frac{1}{2}rv_\perp = \frac{1}{2}rv_\theta = \frac{1}{2}r(\omega r) = \frac{1}{2}r^2\omega = \frac{1}{2}r^2 d\theta/dt = \frac{1}{2}r^2\dot{\theta}$$

Thus angular momentum can also be written as $\boxed{L = mr^2\dot{\theta}}$

METHOD FOR SOLVING FOR ORBITAL TRAJECTORIES IN CENTRAL FORCE MOTION:

Suppose we know the initial conditions in central force motion i.e r_0, θ_0, $\dot{r}_0$ and $\dot{\theta}_0$ of a particle (initial coordinates and radial and tangential velocities), and we wish to solve for the complete trajectory as the motion evolves in future, i.e we wish to know $r(t)$ and $\theta(t)$. To achieve this end we will try to develop a differential equation in one variable at a time which

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can be integrated to solve for the derived variables as a function of time. Let E be the total orbital energy and L be the total angular momentum and $U(r)$ be the potential energy function then:

$$E = U(r) + \frac{1}{2} m\dot{r}^2 + \frac{1}{2} m(r\dot{\theta})^2 \quad (1)$$

And

$$L = mr^2\dot{\theta} \implies \frac{1}{2} m(r\dot{\theta})^2 = L^2/2mr^2 \quad (2)$$

Now equation (1) is a differential equation involving r, θ and t however if we use (2) to substitute last term in (1), suddenly we have a differential equation in a simple variable. Thus using (2) in (1):

$$E = U(r) + \frac{L^2}{2mr^2} + \frac{1}{2} m\dot{r}^2 \quad (3)$$

Now suppose we define a term effective potential energy as:

$$U_{eff}(r) = U(r) + \frac{L^2}{2mr^2}$$ then equation (3) becomes:

$$E = U_{eff}(r) + \frac{1}{2} m\dot{r}^2 \quad (4)$$

Now this problem looks like classical problem of particle moving along x axis with potential energy known as a function of x and total initial energy known (Here r is working as x). Equation (4) is now a simple differential equation which we can integrate to find $r(t)$ (We know r_0 and $\dot{r}_0$ and also E from initial conditions). We can also write:

$$F_{eff} = -\frac{\partial U_{eff}}{\partial r} = m\ddot{r} = m v_r \frac{dv_r}{dr}$$ which will give us

radial velocity as a function of r, which can once again be integrated to get $r(t)$. Once we know $r(t)$ we can use the equation: $L = mr^2\dot{\theta}$ to solve for $\theta(t)$, ∵ After putting $r(t)$, this will be a differential equation only in θ and t.

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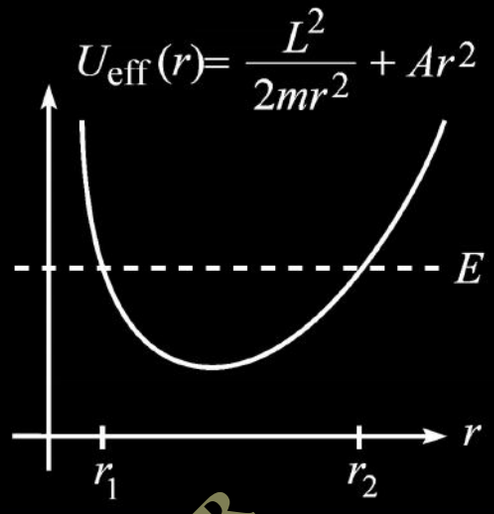
Oscillation Limits Of Orbital Radii :

If a particle is moving in a central field with bounds on maximum and minimum distances from centre of the field, then, it must be trapped in some valley of $U_{eff}(r)$. The procedure for finding minimum and maximum distances from the centre of the field is similar to that for finding bounds for a particle trapped in valley of one dimensional potential energy function. As an example the $U_{eff}(r)$ for the central field $U(r) = Ar^2$ is plotted in the figure. Here E is the total energy of the particle. Where $E = U_{eff}(r)$, the particle's radius will experience a reversal. In the shown figure, r_1 and r_2 represent bounds of the nearest and farthest distances of the orbit. If instead of a valley, particle is in a region around a peak then the orbit may not be stable and particle may spiral out to infinity (Unless it meets a sufficiently high $U_{eff}(r)$ or it may even spiral in towards origin.



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Example : Given L find $U(r)$ that leads to a spiral path of the form $r = r_0 e^{a\theta}$ with $E = 0$. (r_0 and a are constants).

Solution: $E = \frac{1}{2}m\dot{r}^2 + U(r) + \frac{L^2}{2mr^2} = 0$ ①

Now $r = r_0 e^{a\theta} \Rightarrow \dot{r} = r_0 a e^{a\theta}\cdot\dot{\theta} = ar\dot{\theta}$, also $\dot{\theta} = L/mr^2$

$\Rightarrow \dot{r} = aL/mr$. Now, equation ① becomes:

$$0 = \frac{1}{2}m\left(\frac{aL}{mr}\right)^2 + U(r) + \frac{L^2}{2mr^2} \Rightarrow U(r) = \frac{-L^2}{2mr^2}[1+a^2] \text{ Ans.}$$

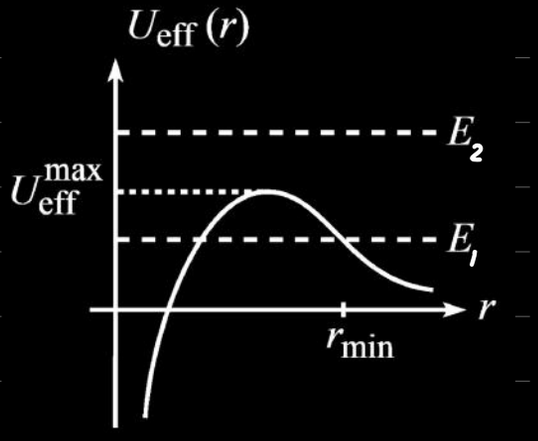
EXAMPLE: A particle moves in a central field such that $U(r) = -C/3r^3$ (a) For a given L what is the maximum possible value of effective potential (b) Suppose particle comes from ∞ with impact parameter b, in terms of C, m and v_0 what is the maximum value of b for which it is captured by the potential (i.e it spirals in to the origin). Here impact parameter means $\perp$ distance to particle's velocity at ∞ from the centre of the field. (v_0 is velocity at ∞).

Solution: $U_{eff} = U(r) + \frac{L^2}{2mr^2} = -\frac{C}{3r^3} + \frac{L^2}{2mr^2}$

for max U_{eff}, $\frac{dU_{eff}}{dr} = 0 \Rightarrow Cr^{-4} - \frac{L^2}{m} r^{-3} = 0$

this gives, $r = mC/L^2$ and corresponding $U_{eff}^{max} = L^6/6m^3C^2$

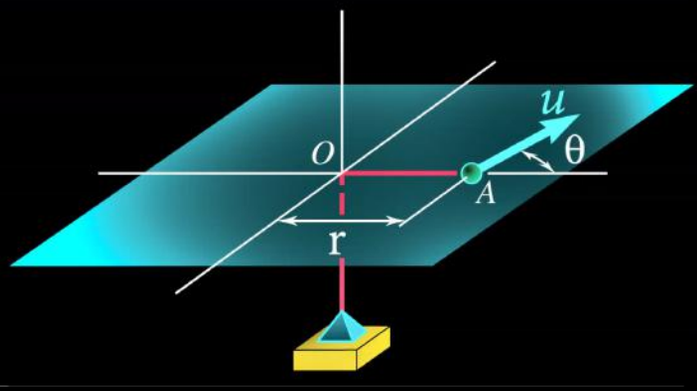
To understand the dynamics, we visualize U_{eff} as shown in the adjacent graph. Here to reach the origin from infinity, particle must be able to overcome the U_{eff}^{max}. Thus if the total energy is E_1, particle will make a reversal at certain r_{min} but if energy is more than U_{eff}^{max} it will be able to reach all the way to the origin.



Now at infinity, $L = mv_0 b$ and $E = \frac{1}{2}mv_0^2$, thus,

$$\frac{1}{2}mv_0^2 > \frac{(mv_0 b)^6}{6m^3C^2} \Rightarrow b < \left(\frac{3C^2}{m^2 v_0^4}\right)^{1/6} \text{ Ans.}$$

EXAMPLE: A particle is connected to a point spring from point O on a smooth horizontal table. The spring is pulled by a distance r and the particle is flicked at an angle θ with radial direction with a speed u. The mass of the particle is m and the



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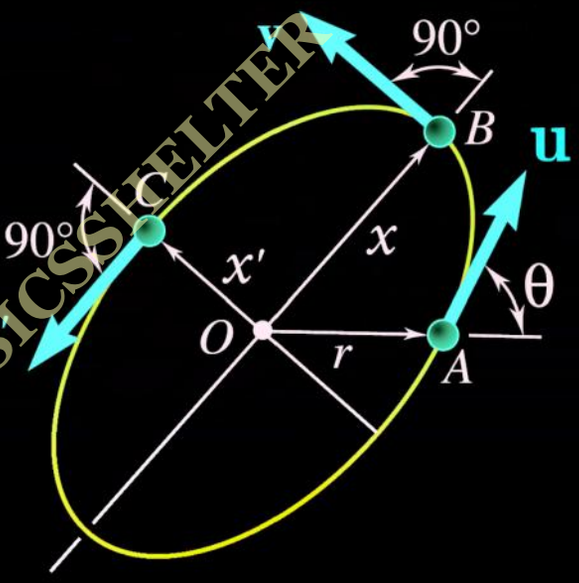
Spring constant is k. Find the max and min distance of the particle from the origin. Given $kr^2/m = u^2$ and $\theta = 60°$

METHOD I: BASICS: Here conservation of mechanical energy and angular momentum will hold. Also at maximum and minimum distances, $\dot{r} = 0$ thus velocity will be perpendicular to the radius. Let x represent extreme distance of the particle from the origin (could be either maximum or minimum distance) and let corresponding speed at the extreme be v. Now



by angular momentum conservation:

$$mur\sin\theta = mvx \quad (1)$$

By energy conservation:

$$\frac{1}{2}kr^2 + \frac{1}{2}mu^2 = \frac{1}{2}kx^2 + \frac{1}{2}mv^2 \quad (2)$$

Now we can eliminate v and solve for x. We will get a quadratic in x^2 from which we will find positive values of x satisfying the equation. The two values represent maximum and minimum distances. Here after putting $kr^2/m = u^2$ and $\theta = 60°$, and solving for x, we get

$$x_{max} = \sqrt{\frac{3}{2}}\,r \text{ and } x_{min} = \frac{r}{\sqrt{2}} \quad \text{Ans.}$$

METHOD II: $E = U_{eff}$: At r_{max} & r_{min} $E = U_{eff} = U(r) + \frac{L^2}{2mr^2}$

Thus: $$\frac{1}{2}kr^2 + \frac{1}{2}mu^2 = \frac{1}{2}kx^2 + \frac{(mur\sin\theta)^2}{2mx^2}$$

Now using $kr^2/m = u^2$ and $\theta = 60°$:

$$mu^2 = \frac{1}{2}mu^2\frac{x^2}{r^2} + \frac{1}{2}mu^2 \cdot \frac{3}{4}\frac{r^2}{x^2}$$

$$\Rightarrow \quad 1 = \frac{1}{2}\frac{x^2}{r^2} + \frac{3}{8}\frac{r^2}{x^2}, \text{ solving we get } x = \frac{r}{\sqrt{2}} \text{ and } x = \sqrt{\frac{3}{2}}\,r$$

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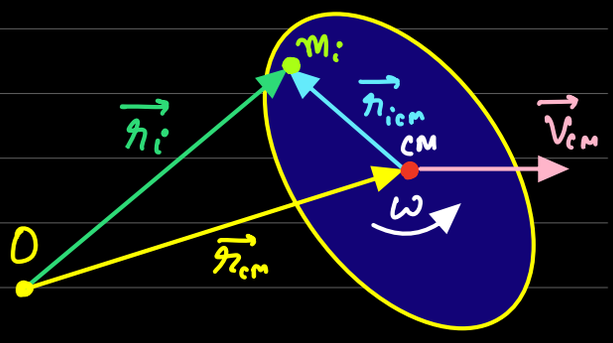
Angular Momentum Of A Rigid Body In General Plane Motion (Axis ⊥ Plane):



Consider a rigid body in general plane motion. Let O be an arbitrary axis perpendicular to the plane of motion about which we wish to write the angular momentum. Let $\vec{r}_{CM}$ be the ⊥ vector to the centroidal axis from O and let $\vec{r}_i$ be the ⊥ vector from O to i^{th} particle and let $\vec{r}_{icm}$ be the ⊥ vector from centroidal axis to i^{th} particle. Now angular momentum of the i^{th} particle about O is given as:

$$\vec{L}_i = \vec{r}_i \times m_i \vec{v}_i = m_i (\vec{r}_{CM} + \vec{r}_{icm}) \times (\vec{v}_{CM} + \vec{v}_{icm})$$

$$= m_i (\vec{r}_{CM} \times \vec{v}_{CM}) + m_i \vec{r}_{icm} \times \vec{v}_{CM} + m_i \vec{r}_{CM} \times \vec{v}_{icm} + m_i \vec{r}_{icm} \times (\vec{\omega} \times \vec{r}_{icm})$$

(I) (II) (III) (IV)

Now for the entire rigid body, we need to sum up the four terms for all particles. We will show that terms II and III become zero.

$$\Sigma(II) = \left(\Sigma m_i \vec{r}_{icm}\right) \times \vec{v}_{CM} = \left(M \vec{r}_{CM,CM}\right) \times \vec{v}_{CM} = 0 \quad (\because \vec{r}_{CM,CM} = 0).$$

$\Sigma III = \vec{r}_{CM} \times \Sigma m_i \vec{v}_{icm}$ but $\Sigma m_i \vec{v}_{icm}$ is momentum in CM frame, which is always zero thus $\Sigma III = 0$

$\Sigma I = (\vec{r}_{CM} \times \vec{v}_{CM}) \Sigma m_i = M (\vec{r}_{CM} \times \vec{v}_{CM})$ and:

$$\Sigma IV = \Sigma m_i \left[\vec{\omega} (\vec{r}_{icm} \cdot \vec{r}_{icm}) - \vec{r}_{icm} (\vec{\omega} \cdot \vec{r}_{icm})\right]$$

$$= \left[\Sigma m_i r_{icm}^2\right] \vec{\omega} - 0 \quad (\because \vec{\omega} \cdot \vec{r}_{icm} = 0 \text{ because } \vec{\omega} \text{ is } \perp \text{ to } \vec{r}_{icm}).$$

$\Rightarrow \Sigma IV = I_{CM} \vec{\omega}$, thus considering all the four terms,

$$\boxed{\vec{L}_O = M (\vec{r}_{CM} \times \vec{v}_{CM}) + I_{CM} \vec{\omega}} \Rightarrow \boxed{\vec{L}_O = \vec{L}_{CM} + \vec{L}}$$

Where $\vec{L}$ is angular momentum about centroidal axis & $\vec{L}_{CM}$ is angular momentum

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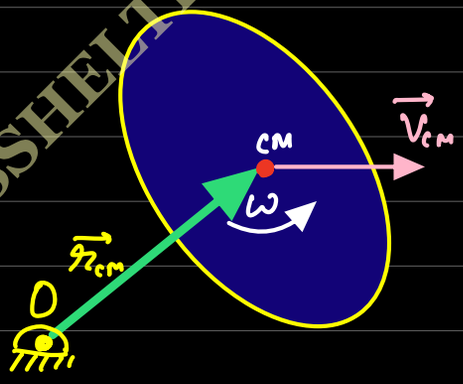
assuming entire mass to be concentrated at centroid.

M: Mass of Rigid body $\vec{r}_{cm}$: ⊥ vector from O to centroidal axis
$\vec{V}_{cm}$: Velocity of centroidal axis, I_{cm}: Moment of Inertia about centroidal axis
$\vec{\omega}$: Angular velocity of the rigid body.

→ Even though angular momentum is a vector quantity, we will treat it like a signed scalar with sign being decided by the terms being clockwise or anticlockwise. The sign of first term is decided using pivot and stick method.

PIVOT & STICK METHOD:

Pivot the stick at O and pin join it to CM now think about the sense of its rotation if it were pulled in direction of $\vec{V}_{cm}$, this sense of rotation is sense of the term $M(\vec{r}_{cm} \times \vec{V}_{cm})$. For instance in adjacent figure, this term is in clockwise sense.

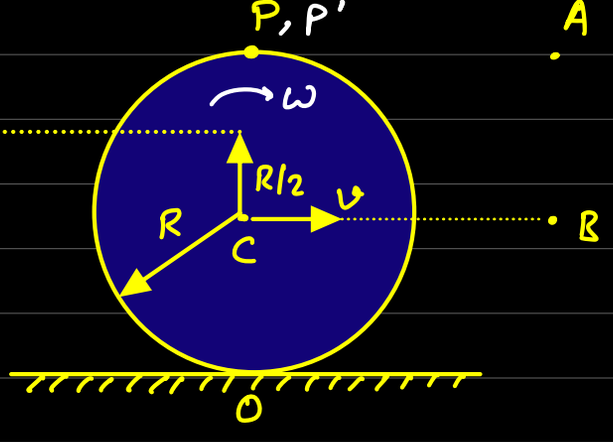


In general we can write signed scalar angular momentum as:

$$L = \pm M(r_\perp V_{cm}) \pm I_{cm}\omega$$

Here ± sign are to be put depending on sense of rotation for instance in the figure shown above if we consider anticlockwise direction as +Ve then, $L_O = -M r_{cm_\perp} V_{cm} + I_{cm}\omega$.

EXAMPLE: For the shown pure rolling wheel find the angular momentum about A, B, D, P (on disc) and P' (fixed in ground frame).



Solution:

(A) $\omega = v/R$ clockwise, using pivot and stick method, $\vec{r}_{cm} \times \vec{V}_{cm}$ term is anticlockwise and $r_\perp = R$ thus, Assuming clockwise as positive

$L = \frac{1}{2}MR^2(v/R) - MRv = \frac{-1}{2}MRv$ Ans.

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(B) In this case $r_\perp = 0$ thus $L = \frac{1}{2}MR^2(v/R) = \frac{1}{2}MRv$ Ans

(D) Here again using pivot & stick method, $\vec{r}_{cm} \times \vec{v}_{cm}$ term is anti-clockwise and $r_{cm\perp} = R/2$ thus, $L = \frac{1}{2}MR^2\left(\frac{v}{R}\right) - M\left(\frac{R}{2}\right)(v) = 0$ Ans.

(P) Be Careful! P is on the disc: From frame of P, cm has a backward velocity ωR (i.e v) thus $\vec{r}_{cm} \times \vec{v}_{cm}$ term is anticlockwise thus $L = \frac{1}{2}MR^2\left(\frac{v}{R}\right) + MRv = \frac{3}{2}MRv$

(Alternatively realize that from frame of P, P itself instantaneous axis of rotation which is as good as permanent axis of rotation for finding angular momentum. Thus $L = I_P\omega = \frac{3}{2}MR^2\omega = \frac{3}{2}MRv$).

(P') P' is fixed on ground as usual, using pivot and stick method $L = \frac{1}{2}MR^2(v/R) - M(R)v = -\frac{1}{2}MRv$

→ Notice that angular momentum is same about P' and A since $r_\perp$ is same for both. Infact if we draw a line parallel to velocity vector then angular momentum about axes through every point on that line $\perp$ to plane of motion will be same ∵ perpendicular distances are same.

→ For any point O on the disc, angular momentum about O is simply $I_O\omega$ ∵ from that frame, O itself becomes instantaneous axis.

Angular Momentum Conservation In 2D Motion:

Recall that about any point or axis we proved that $\vec{\tau} = d\vec{L}/dt$ thus if net torque about some point on axis is zero, then angular momentum about that point on axis will remain conserved. Also recall that in systems, torques of internal forces (action reaction pairs cancel out) thus if net external torque about some axis or point for a system is zero then angular momentum

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of the system about that point remains conserved. Thus:

For a single body if $\vec{\tau} = 0 \Rightarrow \vec{L}$ = Constant

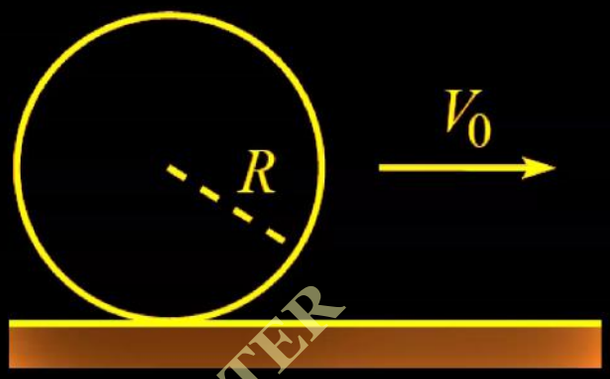
For systems, if $\vec{\tau}_{ext} = 0 \Rightarrow \vec{L}_{sys}$ = Constant.

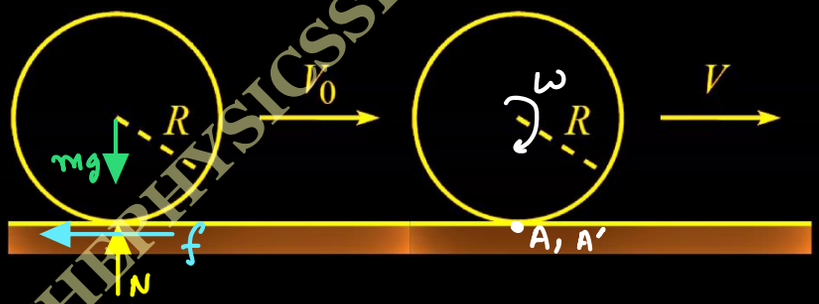
EXAMPLE:

A ball initially slides without rotating on a horizontal surface with friction (see Figure). The initial speed is V_0, and the moment of inertia around the center is $I = \beta m R^2$.

(a) Without knowing anything about the nature of the friction force, find the speed of the ball when it begins to roll without slipping. Also, find the kinetic energy lost while sliding.

(b) Now consider the special case where the coefficient of kinetic friction is μ, independent of position. At what time, and at what distance, does the ball begin to roll without slipping? Verify that the work done by friction equals the energy loss calculated in part (a). (Be careful on this.)





Solution: Here we can see that torque of frictional force about every point of the contact line is zero. Also net torque of N and Mg is zero about every point of the universe. Now let the wheel come to pure rolling at some point A' on the centrode (A is instantaneous axis at that point which coincides with centrode). Since torque about A' is zero, we can conserve angular momentum about that point. Thus:

$$mRV_0 = (\beta mR^2 + mR^2)\omega \Rightarrow \omega R = V = V_0/(1+\beta) \text{ Ans.}$$

(Note that $L_A = L_{A'}$ since velocity of A' is zero relative to A and L_A is simply $I_A\omega$).

Now KE lost : $-\Delta Q = \frac{1}{2}mV_0^2 - \frac{1}{2}\cdot(\beta+1)mR^2\omega^2 = \frac{1}{2}mV_0^2\left[1 - \frac{(\beta+1)}{(1+\beta)^2}\right]$

$\Rightarrow$ KE lost = Heat lost = $\frac{1}{2}mV_0^2\,[\beta/(1+\beta)]$ Ans.

(b) Until pure rolling starts, $a = -\mu g$ thus $V_0 - \mu g T = \frac{V_0}{(1+\beta)}$

$$\Rightarrow T = \frac{V_0}{\mu g}\left[1 - \frac{1}{1+\beta}\right] = \frac{\beta V_0}{(1+\beta)\mu g} \text{ Ans.}$$

Now distance $S = \left(\frac{u+v}{2}\right)\times T = \left(V_0 + \frac{V_0}{1+\beta}\right)\times\frac{1}{2}\times\frac{\beta V_0}{(1+\beta)\mu g}$

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$\Rightarrow \quad S = \frac{\beta(\beta+2)}{(\beta+1)^2}\frac{v_0^2}{2\mu g}$ Ans.

Now speed of contact point at general time t is, $v = v_0 - \mu g t - \alpha R t$

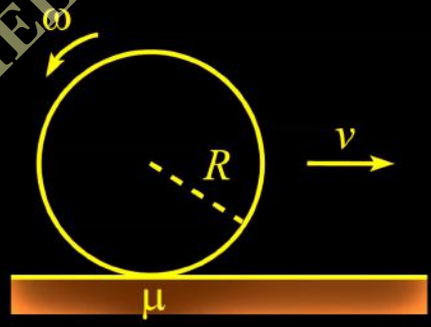
Now $a = f/m$, $\alpha = fR/I_{cm} \Rightarrow \frac{\alpha}{a} = \frac{mR}{I} = \frac{1}{\beta R} \Rightarrow \alpha = \frac{a}{\beta R} = \frac{\mu g}{\beta R}$

$\Rightarrow \quad v = v_0 - \mu g t\left[1 + \frac{1}{\beta}\right]$,

Now work done by friction, $W_f = \int_0^T -\mu mg v\, dt$ $\Rightarrow$ Work done by friction: $W_f = \frac{1}{2}mv_0^2\left(\frac{\beta}{1+\beta}\right)$ Ans.

EXAMPLE :

A coin stands vertically on a table. It is projected forward (in the plane of itself) with speed v and angular speed ω, as shown in Figure. The coefficient of kinetic friction between the coin and the table is μ. What should v and ω be so that the coin comes to rest (both translationally and rotationally) a distance d from where it started?

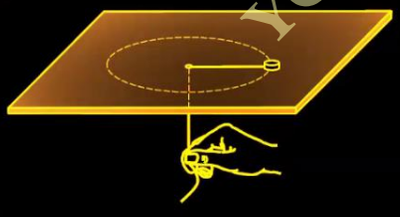


Solution: If finally the coin comes to rest, its angular momentum must be zero about every point ($\because \omega = 0$ & $v_{cm} = 0$). Now conserving angular momentum about arbitrary point on floor, $mvR - \frac{1}{2}mR^2\omega = 0$

$\Rightarrow \quad \omega = 2v/R$ (1) Now $v^2 - u^2 = 2as \Rightarrow 0^2 - v^2 = -2\mu g d$

$\Rightarrow \quad v = \sqrt{2\mu g d}$ Ans and $\omega = \frac{2}{R}\sqrt{2\mu g d}$ Ans.

EXAMPLE: PATHFINDER MCQ 17



A small disc is attached at one end of a light inextensible string that passes through a hole in a frictionless horizontal tabletop. Initially the disc moves on a circle of radius R with kinetic energy K_0. The other end of the string is slowly pulled so that the disc finally moves on a circle of radius R/η. What is the work W done by the pulling agency?

(a) $W = 0$ (b) $W = \eta^2 K_0$

(c) $W = (\eta^2 - 1)K_0$ (d) $W = (\eta - 1)K_0$

Solution: Here the torque of the string tension about the hole is zero and there is no other skew perpendicular force about vertical axis. Thus angular momentum about vertical axis through hole is conserved. Now $K_0 = L^2/2mR^2$ and $K_{final} = L^2/2m(R/\eta)^2$, thus,

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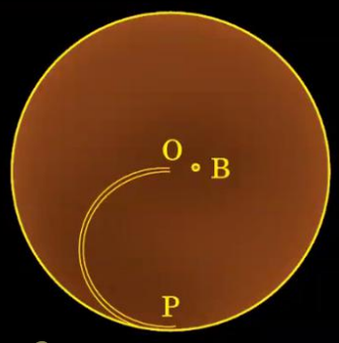
work done by pulling agency $W = K_{final} - K_0 = (\eta^2 - 1) K_0$ Ans.

EXAMPLE: PATHFINDER MCQ 18:

A horizontal disc of mass 1.0 kg mounted on a fixed vertical axle through its centre can rotate without friction. On the disc a semicircular groove OP is cut. A small ball B of mass 1.5 kg when pushed on the disc towards the centre O, the ball enters the groove with a speed 2.0 m/s and moves through the groove without friction. With what speed will the ball leave the disc? Neglect loss of mass due to cutting the groove.



(a) 1.0 m/s
(b) 1.5 m/s
(c) 3.0 m/s
(d) Insufficient information

Solution: Here considering the vertical axis passing through O, there is no external torque. Also, since everything is frictionless, mechanical energy of the system remains conserved. Now initial angular momentum of the system is zero. Let ω be the anticlockwise angular velocity of the disc and v be the speed of the ball at the exit in the ground frame (clockwise angular momentum) then by CAM:

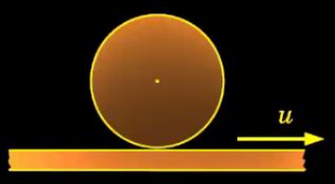
$$0 = \frac{1}{2} \times (1 \text{ kg} \times R^2)\omega - 1.5 \times v \times R \quad \Rightarrow \omega R = 3v \quad (1)$$

Now by COME: $\frac{1}{2} \times 1.5 \times 2^2 = \frac{1}{2} \times 1.5 \times v^2 + \frac{1}{2} \cdot \left(\frac{1}{2} \times 1 \times R^2\right) \omega^2$ (2)

Solving we get: $v = 1 \text{ ms}^{-1}$, $\omega = 3/R$, thus, option (a)

EXAMPLE: PATHFINDER: BUILD-3S:

A uniform wheel of mass m and radius r is free to rotate about a horizontal massless axle that passes through its centre. A horizontal conveyor belt is running underneath the wheel with the help of a motor at a constant velocity u perpendicular to the axle of the wheel. Radius of gyration of the wheel is k.



(a) The wheel is lowered on the belt and then pressed down holding the axle stationary. Find additional energy delivered by the motor during the process of slipping.
(b) The wheel is gently released on the belt. Find additional energy delivered by the motor during the process of slipping.

Solution: Energy supplied must be equal to change in kinetic energy + heat liberated. The heat liberated is frame independent so we can conveniently choose a frame for finding work done by kinetic friction pair.

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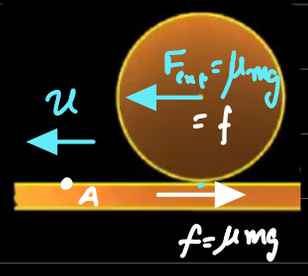
Now let us work from the belt frame:

$\frac{dKE}{dt} = fu + P_f$ where P_f is power developed by friction.



$\frac{dL}{dt} = fr \Rightarrow \frac{dKE}{dt} = \frac{u}{r}\frac{dL}{dt} + P_f$ ①

(Note that angular momentum about A is changing due to external force on the axle).

Now integrating ①, $\Delta KE = \frac{u}{r}\Delta L + W_f$

$\Rightarrow W_f = \Delta KE - \frac{u}{R}\Delta L = \frac{1}{2} \times m(k^2+r^2)\left(\frac{u}{r}\right)^2 - \frac{1}{2}mu^2$

$- \frac{u}{r}\left[m(r^2+k^2)\frac{u}{r} - (mur)\right]$

$\Rightarrow W_f = \frac{1}{2}mu^2\frac{k^2}{r^2} - \frac{u}{r}\left[mk^2\frac{u}{r}\right] = -\frac{1}{2}mu^2\frac{k^2}{r^2}$

Now work done by external agent in ground frame = $\Delta KE - W_f$

$\Rightarrow W_{ext} = \frac{1}{2}mk^2\left(\frac{u}{r}\right)^2 + \frac{1}{2}mu^2\frac{k^2}{r^2} = mu^2\frac{k^2}{r^2}$ Ans.

METHOD 2: USING TIME OF SLIPPING: $\alpha = fr/mk^2$, $\omega = u/r$

$\Rightarrow T = \omega/\alpha = mk^2u/fr^2$

Now to maintain constant speed external agent must also apply a force f on the belt. Thus, power developed by external agent, $P = fu \Rightarrow W_{ext} = \int f u\,dt = fuT = fu \times \frac{mk^2u}{fr^2}$

$\Rightarrow W_{ext} = \frac{mk^2u^2}{r^2}$ Ans. (This was easier).

(b) Once again we will try to find time of slipping. Working from belt frame, by conservation of angular momentum:

$mur = m(k^2+r^2)\omega \Rightarrow \omega = \frac{ur}{r^2+k^2}$, $\alpha = fr/mk^2$

$\Rightarrow T = \omega/\alpha = \frac{mk^2u}{f(r^2+k^2)}$, Now $W_{ext} = fu \times T = \frac{mu^2k^2}{(r^2+k^2)}$ Ans.

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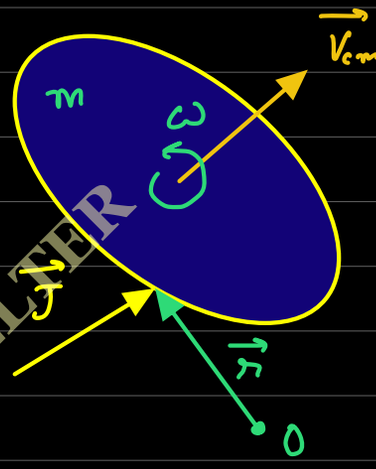
→ Notice the technique in method I of part (a), this can be useful for finding work done by friction sometimes, when a body gets from slipping to pure rolling).

Angular Impulse:

If a sudden impulse $\vec{J}$ acts on a body at a position vector $\vec{r}$ wrt some point O then its angular impulse is defined as:

$$\boxed{\vec{H} = \vec{r} \times \vec{J}}$$ for non impulsive forces,



angular impulse is defined as time integral of torque i.e:

$$\vec{H} = \int \vec{\tau}\, dt = \int \vec{r} \times \vec{F}\, dt$$

Infact first definition is a corollary of the general definition as time integral of torque. When a sudden force is applied for a brief interval of time, $\vec{r}$ may be assumed to be constant and hence can be taken outside the integral and we are left with $\vec{H} = \vec{r} \times \int \vec{F}\, dt = \vec{r} \times \vec{J}$.

Angular Impulse Momentum Theorem:

Net angular impulse acting on a body about any point or axis is equal to the change in angular momentum of the body about that point or axis. This can be readily shown as follows:

$$\vec{\tau} = d\vec{L}/dt \Rightarrow \vec{\tau}\, dt = d\vec{L}$$

$$\Rightarrow \int_0^t \vec{\tau}\, dt = \int_{\vec{L}_i}^{\vec{L}_f} d\vec{L} \Rightarrow \boxed{\vec{H} = \Delta\vec{L}} \quad \text{QED.}$$

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→ While analysing sudden phenomena like collision on tightening of a strings, angular impulse of relatively smaller forces like spring forces or gravity is ignored, though for long term phenomena, we can't ignore them.

Angular Impulse Momentum Theorem For Systems:

Since torques of internal forces cancel out so do the angular impulses of internal torques therefore net change in angular momentum of a system is simply equal to net external angular impulse. i.e.

$$\vec{H}_{ext} = \Delta\vec{L}_{sys}$$

Example (Striking a stick): A stick of mass m and length ℓ, initially at rest, is struck with a hammer. The blow is made perpendicular to the stick at one end. Let the blow occur quickly, so that the stick doesn't have time to move much while the hammer is in contact. If the CM of the stick ends up moving at speed v, what are the velocities of the ends right after the blow?

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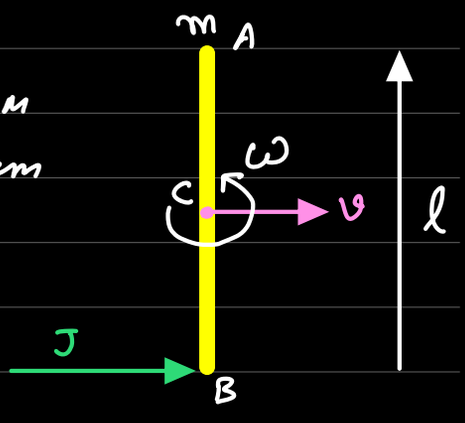
Solution: Let J be the impulse due to hammer. Let C be a point of ground coinciding with CM of the rod, now by Impulse momentum theorem (LIM):

$$J = mv \quad (1)$$

By angular impulse momentum theorem about point C:

$$J\frac{\ell}{2} = \frac{m\ell^2}{12}\omega \quad (2)$$



From (1) & (2), $\omega = 6v/\ell$, thus $v_A = v - \omega\frac{\ell}{2} = -2v$ Ans

$v_B = v + \omega\ell/2 = 4v$ ans.

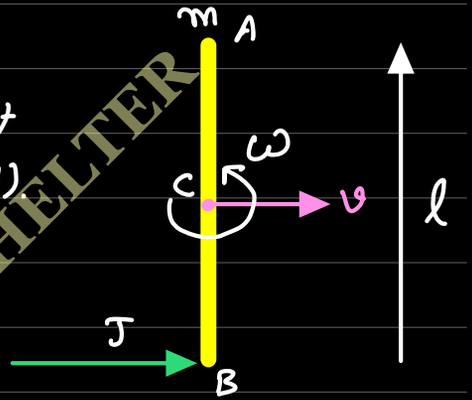
→ Alternatively we can write angular impulse momentum about a point B' on the ground coinciding with B, we get:

$$0 = \frac{m\ell^2}{12}\omega - mv\frac{\ell}{2} \Rightarrow \omega = 6v/\ell \text{ (same result).}$$

PSEUDO IMPULSE: The impulse of pseudo force is called pseudo impulse. Suppose the velocity of certain particle suddenly changes by an amount $\Delta\vec{v}$ then we need to apply a pseudo impulse $-m\Delta\vec{v}$ at the CM of observed body to account for the fact that during the velocity change our frame was non inertial. The pseudo impulse being $-m\Delta\vec{v}$ can be readily shown by integrating pseudo force.

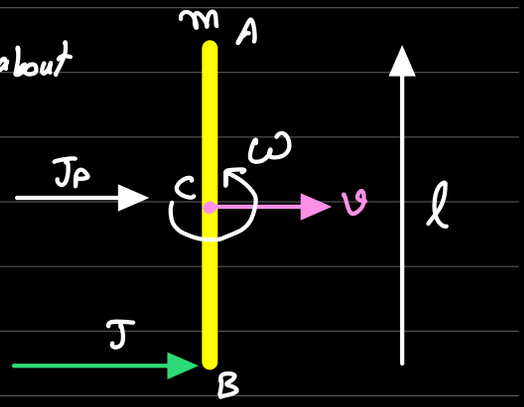
$$\vec{F}_p = -m\vec{a} = -m\frac{d\vec{v}}{dt} \Rightarrow \int \vec{F}_p\,dt = -m\Delta\vec{v}. \quad \text{QED.}$$

EXAMPLE: For the rod of previous example, write the angular impulse momentum equation about end A (particle A is on rod and not ground).



Solution: A will suddenly get a speed $\frac{\omega l}{2} - v$ in the backward direction accordingly we must apply a pseudo impulse on CM. The pseudo impulse, $J_p = m\left(\frac{\omega l}{2} - v\right)$. Now applying AIM about A:

$$m\left(\frac{\omega l}{2} - v\right)\frac{l}{2} + Jl = \frac{ml^2}{3}\omega \quad (3)$$



also $J = mv$ (4) solving (3) & (4) also we get $\omega = 6v/l$.

→ Think! Where is the instantaneous axis? Can you recall the idea of centre of percussion?

Solution: Instantaneous axis can be readily found using $r = v/\omega$ i.e $v/(6v/l)$ i.e $l/6$ above the centre.

Now recall that centre of percussion distance from CM is given by $x_p = \frac{k_{cm}^2}{r_{cm}}$ where k_{cm} is radius of gyration about CM and r_{cm} is distance of CM from pivot. Let y be the distance of instantaneous pivot above CM, thus in our case, $r_{cm} = k_{cm}^2/x_p$

$\Rightarrow r_{cm} = \frac{l^2}{12} / (l/2) = l/6$, this is the same result that we got earlier.

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Conservation Of Angular Momentum (collisions):

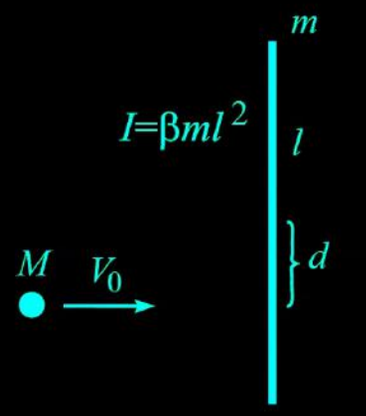
If net external angular impulse about some point in a collision is zero then angular momentum about such a point remains conserved.

→ In case of free collisions the impulses involved are only internal impulse, so we can conserve angular momentum about any axis ⊥ to the plane of motion (Assuming planar motion only).

→ In case one of the bodies is pivoted, angular impulse of the pivot forces about the pivot is zero so angular momentum can be conserved.

Example:

A ball of mass M collides with a stick with moment of inertia $I = \beta m\ell^2$ (relative to its center, which is its CM). The ball is initially traveling at speed V_0 perpendicular to the stick. The ball strikes the stick at a distance d from the center (see Figure). The collision is elastic. Find the resulting translational and rotational speeds of the stick, and also the resulting speed of the ball.



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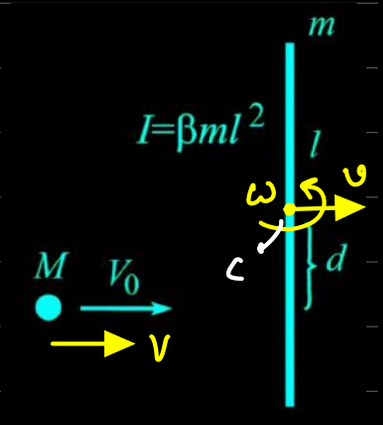
Solution: Let post collision speed of centre of rod be v and that of ball be V and let post collision angular velocity be ω. Then:

CAM_C: $MV_0 d = Mvd + \beta m\ell^2\omega$ ①

COM: $MV_0 = MV + mv$ ②

COME: $\frac{1}{2}MV_0^2 = \frac{1}{2}MV^2 + \frac{1}{2}mv^2 + \frac{1}{2}(\beta m\ell^2)\omega^2$ ③



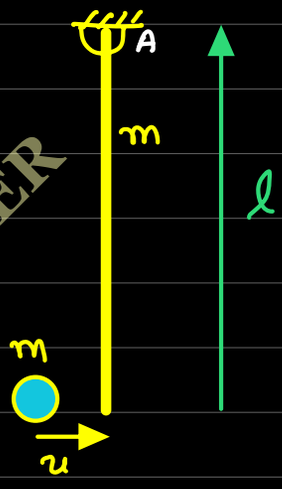
Solving ①, ② & ③ for v, V and ω:

$$V = V_0\,\frac{Md^2 + (M-m)\beta\ell^2}{Md^2 + (M+m)\beta\ell^2},\quad v = \frac{2\beta M\ell^2 V_0}{Md^2 + (M+m)\beta\ell^2},\quad \omega = \frac{2MdV_0}{Md^2 + \beta(M+m)\ell^2}$$

→ One can verify that here velocity of separation is equal to velocity of approach i.e $v + \omega d - V = V_0$

→ This is not only true here but in general whenever there will be elastic collision, velocity of separation will be equal to velocity of separation at point of impact.

EXAMPLE: A ball of mass m hits a pivoted rod as shown in the figure. Find post collision speed of the ball (v), angular velocity of the rod (ω) and pivot impulse during the collision (J_p). Given that collision is perfectly elastic.



Solution:

$$CAM_A: \quad mul = mvl + \frac{ml^2}{3}\omega \quad (1)$$

Restitution: $\omega l - v = u$ (2)

Solving (1) & (2): $v = \frac{u}{2}$, $\omega = 3u/2l$

Now let Pivot impulse be J_p then applying impulse momentum theorem for the system:

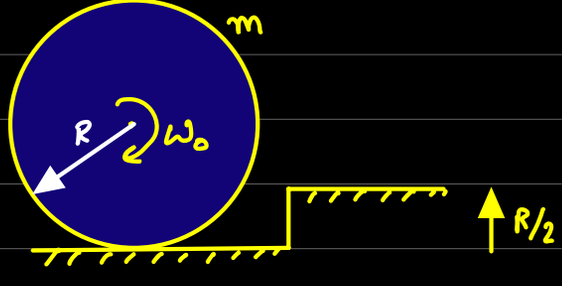
$$J_{ext} = \Delta p_{sys} \Rightarrow J_p = m(v-u) + m\frac{l}{2}\omega$$

$$\Rightarrow J_p = 5mu/4 \text{ Ans.}$$

EXAMPLE: A cylinder of mass m and radius R is rolling without slipping towards a staircase step of height $R/2$, with an angular velocity ω_0. Assume that there is no slipping at the edge of the stair. Find (a) Post impulse ω of the cylinder

(b) Minimum value of ω_0 so as to climb the step

(c) Maximum permissible value of ω_0 so that it does not fly.



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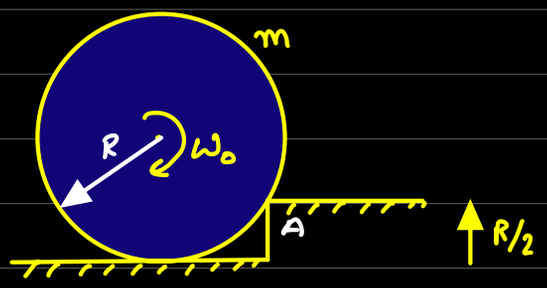
<u>Solution:</u> Just after the collision, edge of the step becomes instantaneous axis and Normal and friction pass through it. Thus we can conserve angular momentum about the step. Thus: CAM_A : $\frac{1}{2}mR^2\omega_0 + m(\omega_0 R)\frac{R}{2} = \frac{3}{2}mR^2\omega$ ①



(Where ω is post collision speed). Thus from ①,
$\omega = 2\omega_0/3$ Ans (a).
Now to climb the step the disc should have enough KE for conversion to PE in the process of climbing the step. Thus:

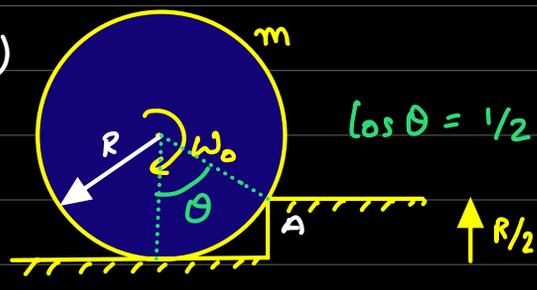
$$\frac{1}{2} \times \frac{3}{2}mR^2\left(\frac{2\omega_0}{3}\right)^2 = mg\frac{R}{2} \quad \text{(In limiting case)}.$$

Thus $\omega_0 \geq \sqrt{3g/2R}$ Ans (b).

Now as the cylinder begins to climb, Normal reaction from the step edge has to balance centrifugal force and component of weight along the normal. (centrifugal force is maximum immediately after the collision and component of weight is least here thus normal reaction will be least immediately after the collision. Now balancing centrifugal force and weight component:

$$m\left(\frac{2\omega_0}{3}\right)^2 R = mg\cos\theta = mg/2 \quad \text{(Limiting case)}$$

$$\Rightarrow \omega_0 \leq \sqrt{\frac{9g}{8R}}$$



→ Note that ω required for climbing the step is more than that for climbing without flying off which means that climbing the step without flying off is not possible. The answer of part (b) is subject to the condition that step could offer some negative normal reaction (through magnetic attraction or some velcro like hook).

→ The wheel will follow a projectile path after flying off.

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3D Rotation About A Point:

Imagine a spherical shell mounted on a circular stand. If we rotate the sphere on the stand a little in any sense, there will always be two diametrically opposite points which will not move under such perturbation of the sphere. For instance if we turn the sphere about a vertical axis through some angle $d\theta$, topmost and bottommost points will be instantaneously at rest (in case of a solid, entire vertical diameter will be at rest). Thus no matter how the sphere is rotated keeping its centre at rest, at every instant there are two diametrically opposite points on its surface whose instantaneous velocity is zero. The line joining these two points forms the instantaneous axis of rotation. With time instantaneous axis may change, however if the centre of the sphere remains fixed, we say that sphere is undergoing rotation about a fixed point. The instantaneous angular velocity is said to be along the diameter on which the points have instantaneous zero velocity.

We can extend this idea to an arbitrary rigid body. If certain point on a rigid body (or its rigid extension) always remains fixed, the rigid body is said to be undergoing rotation about that point. We can always imagine a sphere centred at that point with the actual rigid body occupying a subset of its volume. As the sphere rotates about that centre, the rigid body changes its orientation and the instantaneous axis is defined by diameter of the sphere whose points have instantaneous velocity equal to zero.

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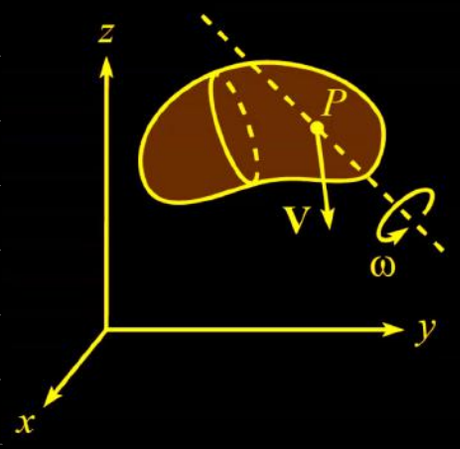
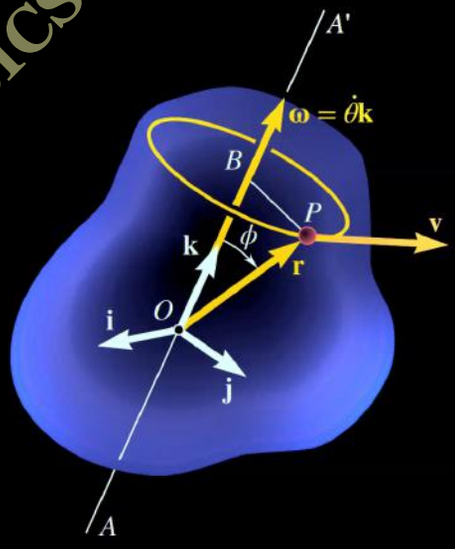
Chasle's Theorem:



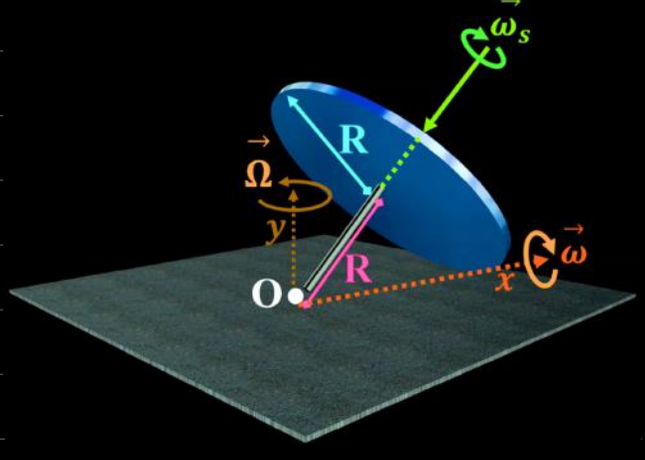
Consider a rigid body undergoing arbitrary motion. Now pick any point P on the body. Then at any instant, the motion of the body can be written as translational motion of P plus a rotation about some axis passing through P (though that axis may change with time). This also means that from point P, the rigid body will alway appear to be rotating about some axis.

Angular Velocity Vector And Velocity Of Arbitrary Point:



Recall that an instantaneous axis is as good as a fixed axis for purpose of finding linear velocity and for fixed axis rotation we showed that velocity vector is related to $\vec{\omega}$ by the relation $\vec{v} = \vec{\omega} \times \vec{r}$, the same idea can also be used for the case of general rotation about a point. i.e. If $\vec{r}$ is position vector of an arbitrary point P wrt some other arbitrary point O then velocity of P wrt O is given as $\vec{v} = \vec{\omega} \times \vec{r}$.
(Why this holds true was proved in the lectures on fixed axis rotation).

Angular Velocity Addition Theorem:



Let S_1, S_2 and S_3 have a common origin. Let S_1 rotate with an angular velocity $\vec{\omega}_{12}$ with respect to S_2 and let S_2 rotate with angular velocity $\vec{\omega}_{23}$ with respect to S_3, then S_1 rotates with respect to S_3 with an instantaneous angular velocity

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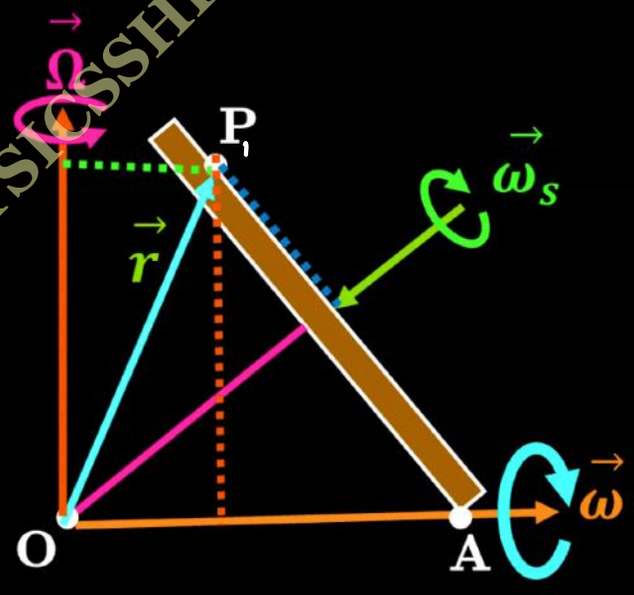
$$\vec{\omega}_{13} = \vec{\omega}_{12} + \vec{\omega}_{23}.$$

For instance consider the 3D rolling disc rotating about point O. Let S_3 represent ground frame and let S_2 represent a precessing frame with angular velocity $\vec{\Omega}$ (from this frame disc appears to be doing fixed axis rotation), and let S_1 be a frame rotating with $\vec{\omega}_s$ in the precessing frame (see figure), then according to this theorem, net angular velocity of S_1 is $\vec{\omega}_s + \vec{\Omega}$. (This is also the net angular velocity of the rolling disc (whose instantaneous axis is horizontal passing through O and contact point of disc with ground).



PROOF: Suppose we pick a point P_1 at rest in S_1. Let $\vec{r}$ be position vector from origin to P_1, then velocity of P_1 (relative to a very close point P_2 at rest in S_2) due to rotation of S_1 around $\vec{\omega}_{12}$ is $\vec{V}_{P_1P_2} = \vec{\omega}_{12} \times \vec{r}$. Now the velocity of point P_2 (relative to a very close point P_3 at rest in S_3) due to rotation of S_2 around $\vec{\omega}_{23}$ is $\vec{V}_{P_2P_3} = \vec{\omega}_{23} \times \vec{r}$ because P_2 is also essentially located at position vector $\vec{r}$. Thus velocity of P_1 relative to P_3 is $\vec{V}_{P_1P_3} = \vec{V}_{P_1P_2} + \vec{V}_{P_2P_3}$

i.e $\vec{V}_{P_1P_3} = \vec{\omega}_{12} \times \vec{r} + \vec{\omega}_{23} \times \vec{r} = (\vec{\omega}_{12} + \vec{\omega}_{23}) \times \vec{r}$. Now this holds for any arbitrary point P_1 at rest in S_1 thus frame S_1 rotates with angular velocity $\vec{\omega}_{12} + \vec{\omega}_{23}$ wrt S_3.

→ The proof essentially follows from facts that (1) linear velocities add as usual and (2) Angular velocities differ from linear velocities by a cross product with $\vec{r}$.

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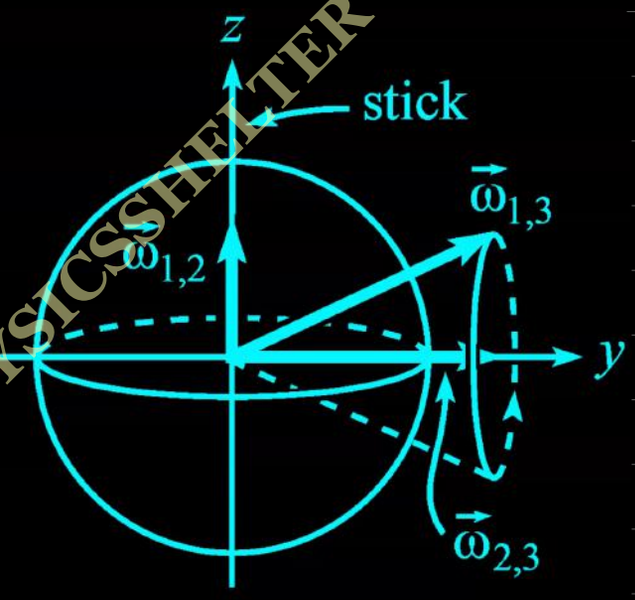
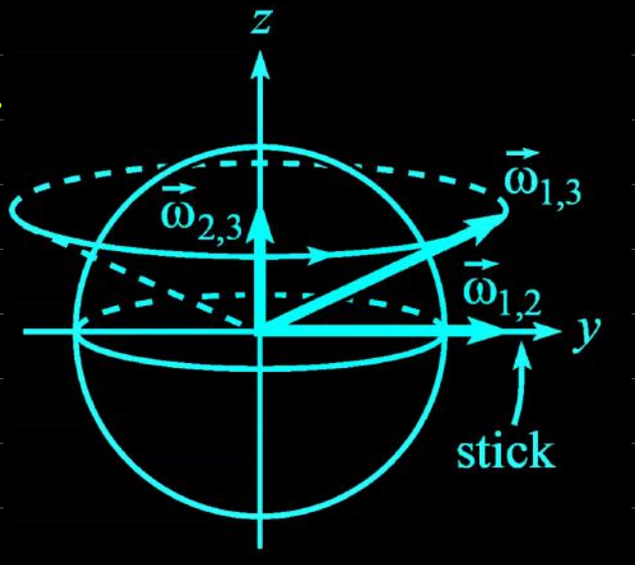
→ Thus with our rolling disc example, net angular velocity of the disc is $\vec{\omega} = \vec{\Omega} + \vec{\omega}_s$ where $\vec{\omega}_s$ is the angular velocity of spinning about an axis fixed in the frame precessing with angular velocity $\vec{\Omega}$ wrt ground.

Example (Rotating sphere): A sphere rotates with angular speed $\vec{\omega}_3$ around a stick that initially points in the $\hat{z}$ direction. You grab the stick and rotate it around the $\hat{y}$ axis with angular speed $\vec{\omega}_2$. What is the angular velocity of the sphere, with respect to the lab frame, as time goes by?

Solution: In the language of angular velocity addition theorem, the sphere is the S_1 frame, the stick and $\hat{y}$ define S_2 frame and lab is the S_3 frame. (spinning is happening about z axis and spinning axis is precessing about y axis). Thus $\vec{\omega}_{12} = \omega_3 \hat{z}$ and $\vec{\omega}_{23} = \omega_2 \hat{y}$ thus according to the theorem, $\vec{\omega}_{13} = \vec{\omega}_{12} + \vec{\omega}_{23}$ $= \omega_3 \hat{z} + \omega_2 \hat{y}$ (This is shown in the figure). Note that as the spinning axis precesses, $\vec{\omega}_{13}$ traces a cone about y axis.



→ If spinning were happening about y axis initially and spinning axis were to precess about z axis, then net $\vec{\omega}$ of sphere $\vec{\omega}_{13}$ would trace a cone about z axis as shown in adjacent figure.



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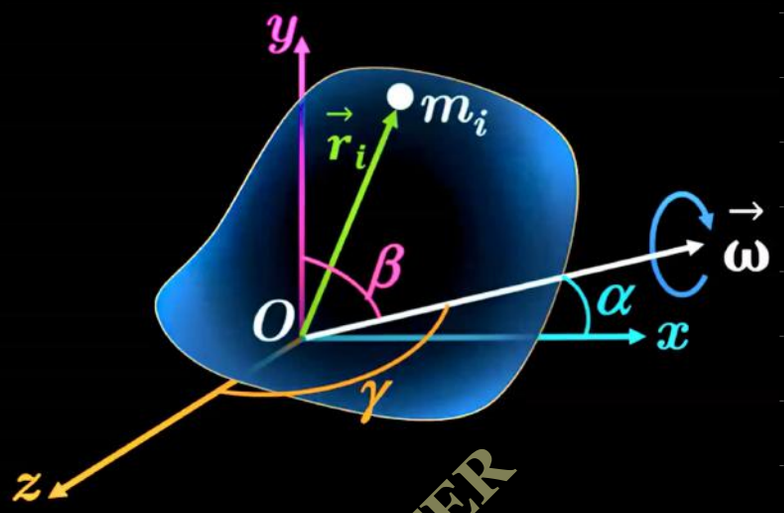
Angular Momentum Of A Rigid Body About A Point:

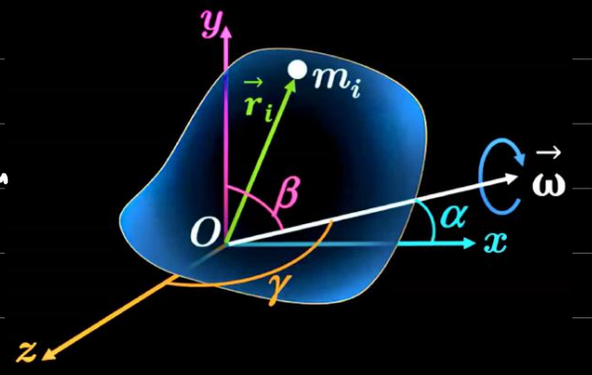
Consider a rigid body rotating with angular velocity $\vec{\omega}$ about point O. We wish to find angular momentum of the body about point O. (See figure). We will prove that x component of angular momentum is given as:



$$L_x = I_x \omega_x - I_{xy} \omega_y - I_{xz} \omega_z$$

and similar cyclic expressions hold for y & z directions. Here I_{xy} and I_{xz} are products of inertia (defined in lectures on fixed axis rotation). ω_x, ω_y and ω_z are the respective components of angular velocity vector.

Proof: Consider a general particle of mass m_i located at position vector $\vec{r}_i$. By definition of angular momentum of a particle about a point, its angular momentum about O is given as:



$$\vec{L}_i = m_i \vec{r}_i \times (\vec{\omega} \times \vec{r}_i)$$

$$= m_i (\vec{\omega}(\vec{r}_i \cdot \vec{r}_i) - \vec{r}_i (\vec{\omega} \cdot \vec{r}_i))$$

(∵ Vector triple product formula: $\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b}(\vec{a} \cdot \vec{c}) - \vec{c}(\vec{a} \cdot \vec{b})$).

$$\Rightarrow \vec{L}_i = m_i [\vec{\omega} r_i^2 - \vec{r}_i (\vec{\omega} \cdot \vec{r}_i)] \quad (1)$$

Now $r_i^2 = x_i^2 + y_i^2 + z_i^2$ and $\vec{\omega} = \omega_x \hat{i} + \omega_y \hat{j} + \omega_z \hat{k}$ and $\vec{r}_i = [x_i \hat{i} + y_i \hat{j} + z_i \hat{k}]$

Substituting in equation (1):

$$\vec{L}_i = m_i (x_i^2 + y_i^2 + z_i^2) [\underline{\omega_x \hat{i}} + \omega_y \hat{j} + \omega_z \hat{k}]$$

$$- m_i (\underline{x_i \hat{i}} + y_i \hat{j} + z_i \hat{k}) (x_i \omega_x + y_i \omega_y + z_i \omega_z) \quad (2)$$

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Here the terms contributing to x component of angular momentum have been written in green color and underlined. We will collect the terms contributing to x component of angular momentum and write y and z components by symmetry (cyclic exchange)).

Now collecting coefficients of $\hat{i}$ in equation (2):

$$L_{ix} = m_i (y_i^2 + z_i^2)\omega_x - m_i x_i y_i \omega_y - m_i x_i z_i \omega_z$$

Now upon summation, for the entire rigid body we set:

$$\boxed{L_x = I_x \omega_x - I_{xy}\omega_y - I_{xz}\omega_z}$$ Q.E.D.

If suppose we had chosen x, y and z axes as principal axes then, $I_{xy} = 0$ and $I_{xz} = 0$ and we are left with:

$L_x = I_x \omega_x$ and thus by symmetry:

$$\boxed{\vec{L} = I_x \omega_x \hat{i} + I_y \omega_y \hat{j} + I_z \omega_z \hat{k}} \quad (3)$$

← Angular momentum in rotation about fixed point using principal axes.

→ In case of general motion, we break it in to translation of CM and rotation about CM and our formula becomes:

$$\boxed{\vec{L} = M(\vec{r}_{CM} \times \vec{v}_{CM}) + I_x \omega_x \hat{i} + I_y \omega_y \hat{j} + I_z \omega_z \hat{k}}$$

Here I_x, I_y and I_z will be principal moments about centroidal axes.

→ Once we know $\vec{L}(t)$ we can use $\vec{\tau} = d\vec{L}/dt$ about point of rotation to analyze the dynamics of the problem.

→ Angular momentum vector and angular velocity vector need not be in same direction as evident from equation (3) unless, I_x, I_y and I_z are all equal to each other. In such a scenario, the angular velocity vector may change even if there is no external torque (unless body is rotating about

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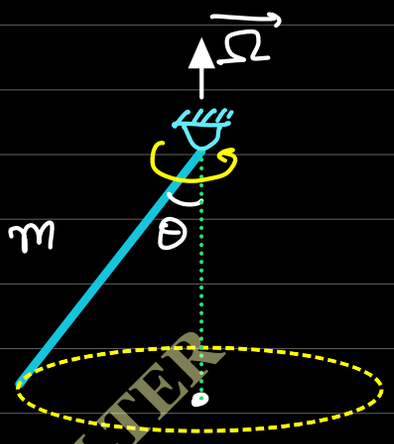
one of the principal axes). This is because in absence of torque, angular momentum must be conserved even if the angular velocity needs to be changed to keep angular momentum constant.

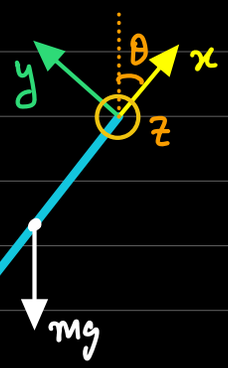
EXAMPLE: IRODOV: 1·270:

A thin uniform rod of mass m and length l rotates uniformly about a vertical axis with angular velocity Ω as shown. Find the angle θ between the rod and the vertical.



Solution: **METHOD I: USING PRINCIPAL AXES:**

Consider the adjacent figure. Notice that a slender rod is an example of body with cylindrical symmetry. Thus x, y and z axes shown in the figure are principal axes passing through the top end.



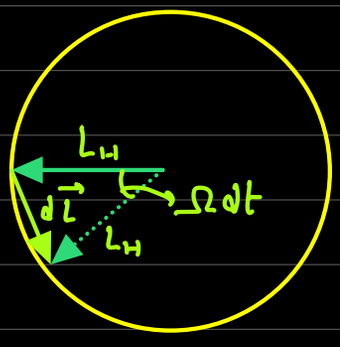
Now:

$\vec{L} = I_x \omega_x \hat{i} + I_y \omega_y \hat{j} + I_z \omega_z \hat{k}$. For our rod, at the shown instants, $\Omega_x = \Omega \cos\theta$, $\Omega_y = \Omega \sin\theta$ and $\Omega_z = 0$.

and, $I_x = 0$, $I_y = ml^2/3$ and $I_z = ml^2/3$ thus:

$$\vec{L} = \frac{ml^2}{3} \Omega \sin\theta \, \hat{j}$$

. Now as the rod rotates vertical component of this angular momentum remains constant but its horizontal component ($L_H = \frac{ml^2}{3} \Omega \sin\theta \cos\theta$)



rotates with angular velocity Ω (see circle diagram).

Thus, change in angular momentum in a time dt:

$$dL = L_H \Omega \, dt \implies \frac{dL}{dt} = L_H \Omega$$

Now torque of gravity about pivot: $\tau_G = mg \frac{l}{2} \sin\theta$

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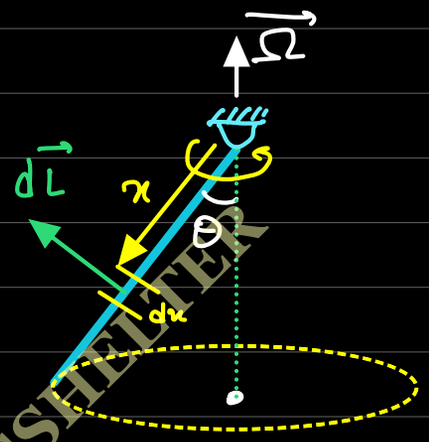
Now using $\vec{\tau} = \frac{d\vec{L}}{dt}$: $mg\frac{l}{2}\sin\theta = \frac{ml^2}{3}\Omega\sin\theta\cos\theta \times \Omega$

$\Rightarrow \cos\theta = \frac{3g}{2\Omega^2 l}$ Ans.

METHOD II: FINDING ANGULAR MOMENTUM BY DIRECT INTEGRATION:



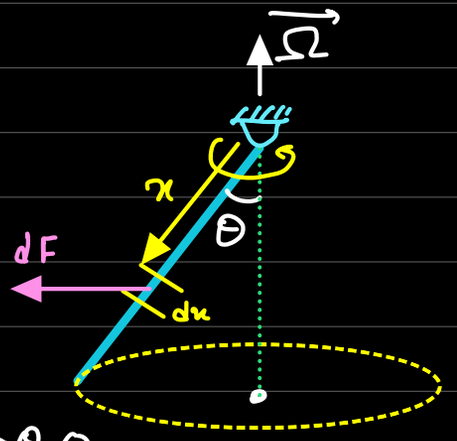
Consider an element of thickness dx at a distance x from the pivot as shown. The velocity of the element is $\Omega x\sin\theta\odot$

Now $d\vec{L} = \vec{x}\times m\vec{v} = \frac{m}{l}dx.\,\Omega x\sin\theta \times x$

$\Rightarrow \int dL = \frac{m\Omega\sin\theta}{l}\int_0^l x^2 dx = \frac{ml^2}{3}\Omega\sin\theta$

Thus $\vec{L} = \frac{ml^2}{3}\Omega\sin\theta$ ← This is the same result that we obtained using principal axes in method I. After this rest of the steps are exactly like those in method I.

METHOD III BALANCING THE CENTRIFUGAL AND WEIGHT TORQUES.



In a rotating frame about vertical axis, the rod appears stationary and hence has constant angular momentum. Thus in this frame, torque of gravity must exactly balance the torque due to centrifugal forces.

$$\int d\tau_c = \int_0^l \frac{m}{l}dx\,\Omega^2 x\sin\theta\cdot x\cos\theta = \frac{m\Omega^2 l^2}{3}\sin\theta\cos\theta \quad (1)$$

τ_c: Centrifugal torque

Torque of gravity: $\tau_g = mg\frac{l}{2}\sin\theta$ (2)

From (1) & (2): $\frac{m\Omega^2 l^2}{3}\sin\theta\cos\theta = mg\frac{l}{2}\sin\theta \Rightarrow \cos\theta = 3g/2\Omega^2 l$ Ans.

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Kinetic Energy In Rotation About A Point:

In case of rotation about a fixed point, we can show that Kinetic Energy is given as $\boxed{KE = \frac{1}{2}\vec{L}\cdot\vec{\omega}}$ This can be shown as follows:

Proof:

$$\vec{L_i} = m_i(\vec{r_i} \times (\vec{\omega} \times \vec{r_i})) = m_i(\vec{\omega} r_i^2 - \vec{r_i}(\vec{\omega}\cdot\vec{r_i})), \text{ thus:}$$

$$\frac{1}{2}\vec{L_i}\cdot\vec{\omega} = \frac{1}{2}m_i[\omega^2 r_i^2 - (\vec{\omega}\cdot\vec{r_i})^2] = \frac{1}{2}m_i\omega^2 r_i^2[1-\cos^2\theta]$$

$$= \frac{1}{2}m_i\omega^2 r_i^2 \sin^2\theta = \frac{1}{2}m_i(\vec{\omega}\times\vec{r_i})^2 = \frac{1}{2}m_i v_i^2 = KE_i$$

Thus: $\sum \frac{1}{2}\vec{L_i}\cdot\vec{\omega} = \frac{1}{2}\vec{L}\cdot\vec{\omega} = \frac{1}{2}\sum m_i v_i^2 = KE$ thus:

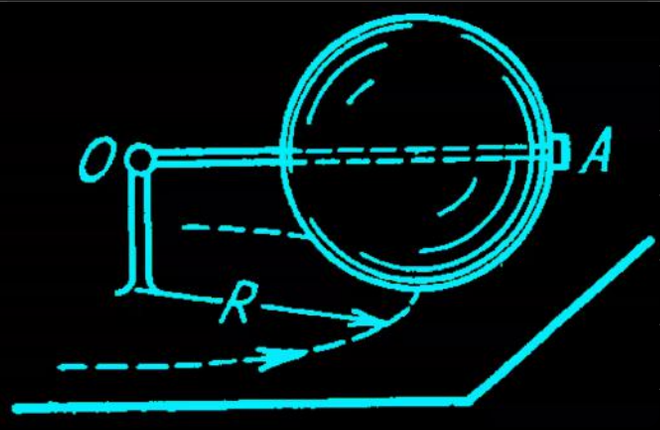
$$\boxed{KE = \frac{1}{2}\vec{L}\cdot\vec{\omega}} \quad \text{QED}$$

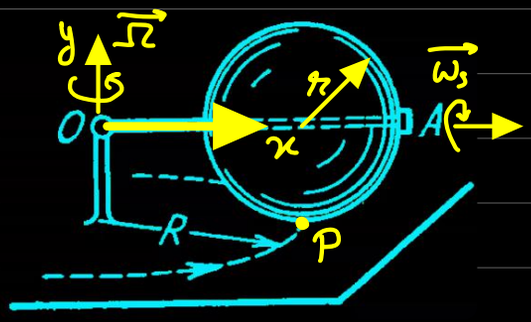
→ In case of general motion we can consider it as superposition of motion about CM and motion of CM. Thus KE becomes:

$$\boxed{KE = \frac{1}{2}Mv_{CM}^2 + \frac{1}{2}\vec{L}_{CM}\cdot\vec{\omega}}$$ $\vec{L}_{CM}$: Angular momentum about CM.

Example: JEE Adv 1.267:

A uniform sphere of mass m and radius r rolls without sliding over a horizontal plane rotating about a horizontal axle OA (see figure). In the process, the centre of the sphere moves with velocity v along a circle of radius R. Find the kinetic energy of the sphere.





Solution: The precession and spinning angular velocities of the sphere are shown in the figure. Since the velocity of point P is zero (pure rolling), $\Omega R = \omega_s r$ ①. Also: $\vec{\omega} = \vec{\Omega} + \vec{\omega}_s$

$\Rightarrow \vec{\omega} = \Omega\hat{j} - \omega_s\hat{i}$. This is a case of rotation about point O. Thus:

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$\vec{L}_o = I_x \omega_x \hat{i} + I_y \omega_y \hat{j} + I_z \omega_z \hat{k}$, Also $\Omega = \frac{v}{R}$ and $\omega_s = \frac{v}{r}$ ($\because$ ①).

Thus $\vec{L}_o = \frac{2}{5} m r^2 \times \frac{v}{r} (-\hat{i}) + \left(\frac{2}{5} m r^2 + m R^2\right) \frac{v}{R} \hat{j}$

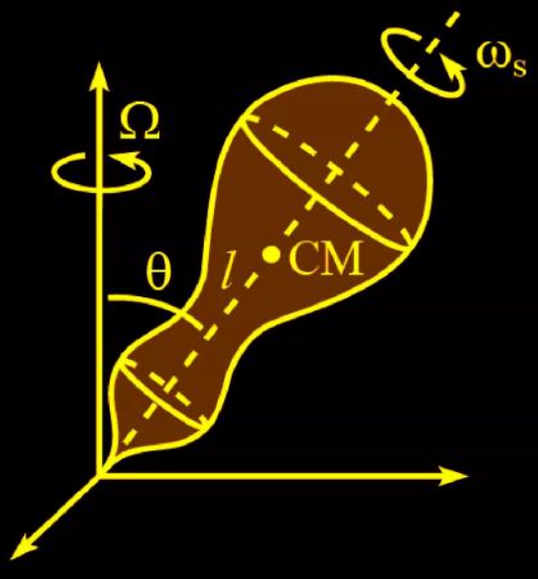
$\vec{\omega} = -\frac{v}{r} \hat{i} + \frac{v}{R} \hat{j}$ thus $KE = \frac{1}{2} \vec{L} \cdot \vec{\omega} = \frac{1}{5} m v^2 + \frac{v^2}{2R^2} \left[\frac{2}{5} m r^2 + m R^2\right]$

$= \frac{7}{10} m v^2 + \frac{1}{5} m v^2 \frac{r^2}{R^2} = \frac{7}{10} m v^2 \left[1 + \frac{2}{7} \frac{r^2}{R^2}\right]$ Ans.

→ Method 2: Alternatively recognize that instantaneous axis passes through O and P and ω is $v/$(⊥ distance of centre from OP). Now KE is simply $\frac{1}{2} I_{OP} \times \omega^2$.

Precession Of A Top:

Example (Morin): A symmetric top of mass m has its CM at a distance l from its pivot. The moment of inertia about the spinning axis is I. The top spins with an angular velocity ω_s about spinning axis and initial conditions are set so that the CM precesses in a circle around the vertical axis. The symmetry axis makes a constant angle θ with the vertical (see figure). Assuming that angular momentum due to ω_s is much larger than any other angular momentum, find an approximate expression for the frequency Ω of the precession.



Solution: Since $\Omega << \omega_s$ (Given angular momentum due to precession is negligible), angular momentum can be written as: $L \approx I\omega_s$. This points along spinning axis (nearly). Now horizontal component of its angular momentum precesses with angular velocity Ω while vertical component remains constant. To visualize the rate of change of angular momentum, we draw a circle of radius L_H i.e horizontal component of angular momentum (i.e $L \sin\theta$).

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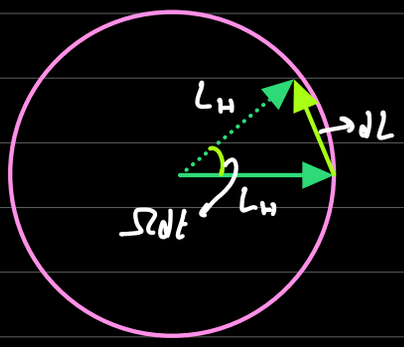
From the circle diagram: $dL = L_H \Omega dt = L \sin\theta \Omega dt$

$\Rightarrow \frac{dL}{dt} = \Omega L \sin\theta = I \Omega \omega_s \sin\theta = \tau_G$



where τ_G is torque due to gravity about the pivot point.

Thus: $I \Omega \omega_s \sin\theta = mg l \sin\theta$, thus,

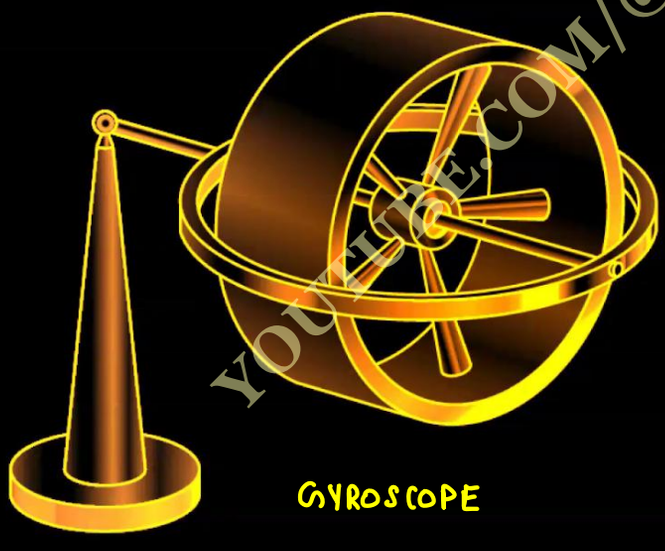
$$\boxed{\Omega = \frac{mgl}{I\omega_s}}$$ ← Precession angular frequency of the top ($\Omega << \omega_s$).

→ The above expression explains why tops begin to precess rapidly as their spinning angular velocity (ω_s) reduces.

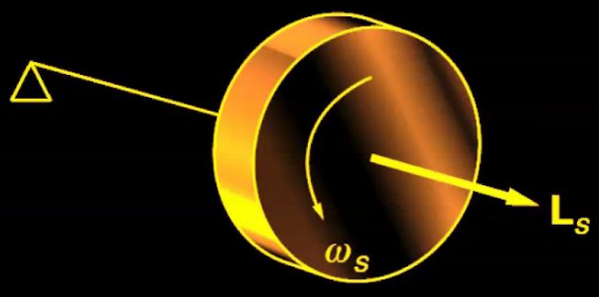
→ Frictional force from ground must provide necessary centripetal force for CM to go in a circle. Thus there may be a limit to θ for which the top may perform fixed point rotation.

→ We have done approximate analysis assuming $\omega_s >> \Omega$. An exact analysis can also be done using our exact formula for angular momentum though it will increase the required bull work considerably.

GYROSCOPE:



GYROSCOPE



SCHEMATIC

A gyroscope involves a flywheel that spins on an axle and a suspension that allows the axle to assume any orientation. The general analysis of gyroscopic motion can get mathematically quite complicated however here we will be mainly concerned with uniform precession of the gyroscope. Just like the case of a top, we can find the

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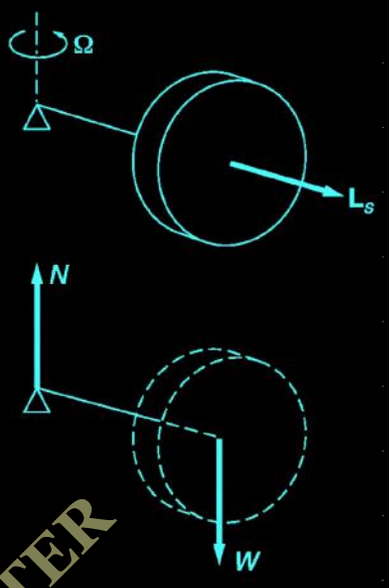
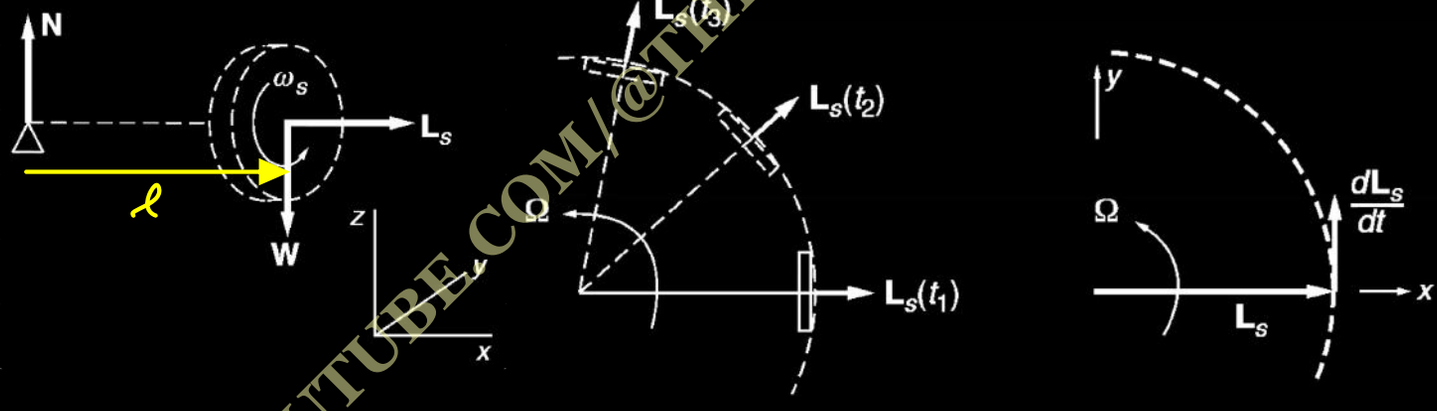
relation between spinning angular velocity and precession angular velocity of the gyroscope. The right figure above is a schematic of the gyroscope. The flywheel is initially brought up to a speed spinning at a rate ω_s usually by pulling a string that is wound around the axle. If the gyroscope is then released from a horizontal position, it wobbles (nutates) briefly and then settles in to a steady precession in which the axle rotates around a vertical axis with a constant angular velocity Ω. Here it might appear strange that gyroscope does not fall, however this can be easily understood in terms of the equation $\tau = dL/dt$. If the wheel were not spinning then the torque of gravity would make the wheel fall however, one can see that torque of gravity is perpendicular to the axle and thus $\vec{\tau} = \frac{d\vec{L}}{dt}$ can be satisfied by precession of the axle.





The above figure are drawn to visualize the process in detail. The phenomenon is very similar to precession of a top. Now torque due to weight of the wheel, $\tau = Wl$ now if L_s is spin angular momentum, then change in angular momentum in a time dt will be $dL = L_s \Omega dt \Rightarrow \frac{dL}{dt} = L_s \Omega = I_0 \omega_s \Omega$

where I_0 is moment of inertia about spinning axis. (Note that unlike spinning top here there is no approximation in dL because the axle here is exactly $\perp$ to precession axis). Thus using $\tau = dL/dt$:

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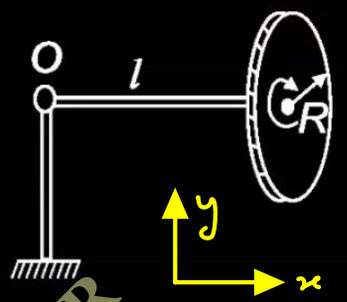
$$Wl = I_0 \omega_s \Omega \implies \boxed{\Omega = \frac{Wl}{I_0 \omega_s}}$$ ← Gyroscope precession frequency.

Example: Irodov 1.284:

A gyroscope, a uniform disc of radius $R = 5.0$ cm at the end of a rod of length $l = 10$ cm (Fig. 1.73), is mounted on the floor of an elevator car going up with a constant acceleration $w = 2.0\ \text{ms}^{-2}$. The other end of the rod is hinged at the point O. The gyroscope precesses with an angular velocity $n = 0.5$ rps. Neglecting the friction and the mass of the rod, find the proper angular velocity of the disc.



rps: revolutions / second.

Solution: $g_{eff} = g + w$, Now as discussed in previous section,

$$\Omega = \frac{M(g+w)l}{\frac{1}{2}MR^2\omega_s} = 2\pi n \quad (\because W = Mg, \text{ and } I_0 = \tfrac{1}{2}MR^2, g_{eff} = g+w)$$

$$\implies \omega_s = \frac{(g+w)l}{\pi R^2 n} = 300\ \text{rad s}^{-1}$$ ← This is given answer in Irodov

However proper angular velocity must include precession Ω also thus answer should be $\vec{\omega} = \vec{\Omega} + \vec{\omega}_s = -300\hat{i} - \pi\hat{j}$ $(\because 2\pi n = \pi)$.

Example: Irodov 1.288:

A ship moves with velocity $v = 36$ km per hour along an arc of a circle of radius $R = 200$ m. Find the moment of the gyroscopic forces exerted on the bearings by the shaft with a flywheel whose moment of inertia relative to the rotation axis equals $I = 3.8 \times 10^3\ \text{kg m}^2$ and whose rotation velocity $n = 300$ rpm. The rotation axis is oriented along the length of the ship.

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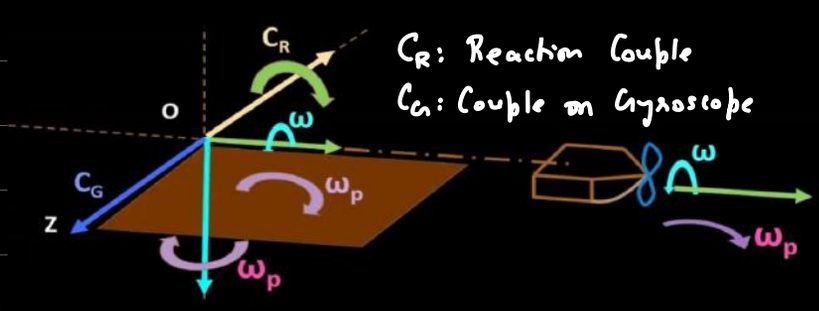
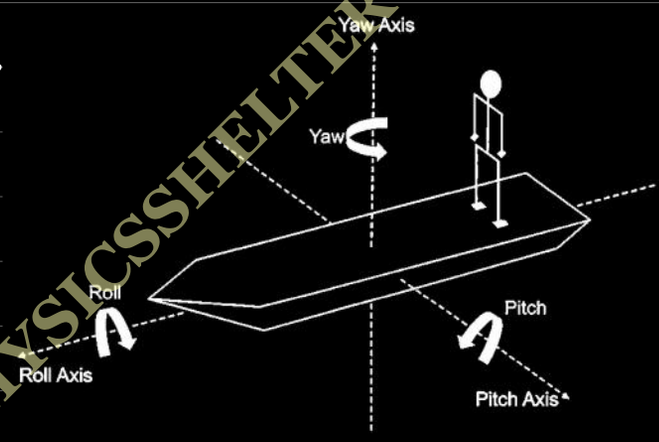
Solution: When ship is taking a turn, its propeller gets a precession Ω in addition to its spinning ω_s about its own axis. This requires a torque to change the angular momentum of the propeller and the reaction torque acts on the ship.



Because of the reaction torque the ship has a tendency to pitch (i.e rotate about the pitch axis).

For instance in the figure show above, the angular momentum of propeller is changing in $+z$ direction so torque on it is in $+z$ direction accordingly, its reaction torque is in $-z$ direction. Thus the body of ship will pitch such that its nose will haven tendency of dipping down and tail will have a tendency of going upwards.

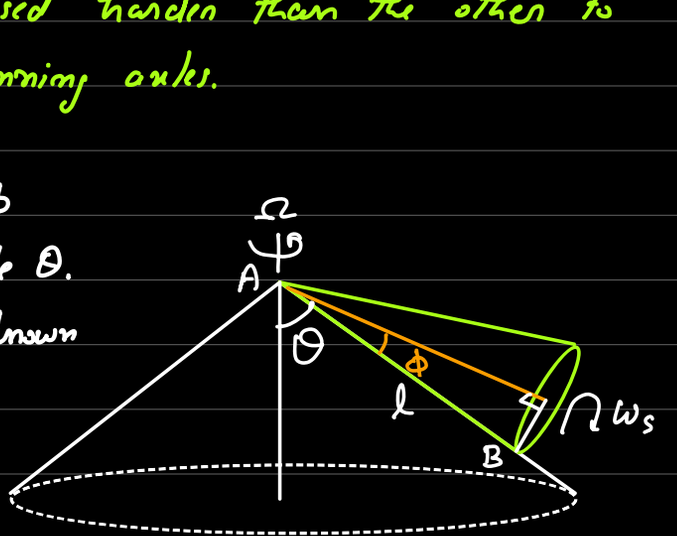


In our case, $\Omega = v/R$. Now required torque is therefore,

$$\tau = I\,\Omega\,\omega_s = I\,\frac{v}{R}\times\frac{2\pi n}{60} = \frac{\pi I v n}{30R} = 5.97\ \text{kNm}$$ Ans.

→ Similar phenomenon also occurs with turbine shafts in aircrafts and Railway locomotive shafts. When the train takes a turn, one of the rails is pressed harder than the other to account for precession of the spinning axles.

EXAMPLE: A cone of semivertex angle ϕ rolls on another cone of semivertex angle θ. If the spinning ω_s of rolling cone is known find its precession Ω.



Solution: Now $\vec{\omega} = \vec{\Omega} + \vec{\omega_s}$. Now net $\vec{\omega}$ is along AB ∴ AB is instantaneous axis of rolling thus components of $\vec{\omega_s}$ and $\vec{\Omega}$ ⊥ to this direction must cancel. Thus $\omega_s \sin\phi = \Omega \sin\theta$

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$\Rightarrow \Omega = \omega_s \sin\phi / \sin\theta$ Ans.

→ Think: In the previous problem, what is the angular acceleration of the rolling cone.

Solution: Here horizontal component of ω_s processes with Ω thus rate of change of angular velocity is simply $\omega_{s_H} \times \Omega = \omega_s (\sin(\theta+\phi))\Omega$

Thus, $\alpha = \frac{\omega_s \sin\theta}{\sin\phi} \times \omega_s \sin(\theta+\phi) = \omega_s^2 \frac{\sin(\theta+\phi)\sin\theta}{\sin\phi}$

→ Think: What is the net ω in previous example

Solution: $\omega = [\omega_s^2 + \Omega^2 + 2\omega_s \Omega \cos(\theta+\phi)]^{1/2}$

→ Think: How will you find KE of rolling cone.

Solution: $KE = \frac{1}{2} I \omega^2$ where I is moment of inertia about slant height (Which can be obtained using $I_u \omega^2\alpha + I_y \omega^2\beta + I_z \omega^2\gamma$ formula). Alternatively we could use $KE = \frac{1}{2} \vec{L} \cdot \vec{\omega}$

→ Think: How will you find normal reaction and friction on the rolling cone?

Solution: CM of cone is undergoing a circular motion with angular velocity Ω. If we know the location of CM we know its centripetal acceleration. Now we can apply $F_{ext} = ma$ along slant height and Normal to evaluate friction and normal (Hence also minimum required value of coefficient of friction).

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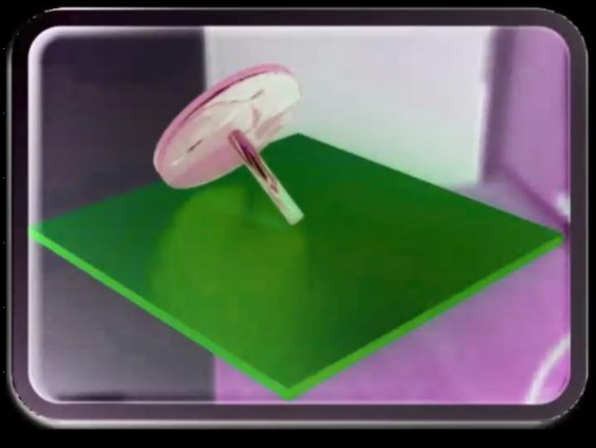
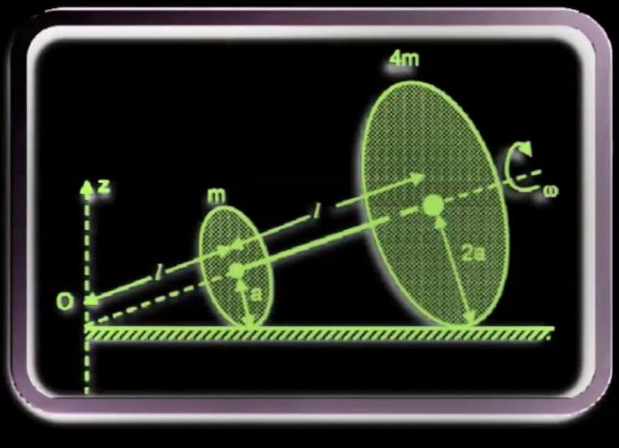
EXAMPLE: JEE ADVANCED 2016:

Problem Statement: Two thin circular discs of mass m and 4m, having radii of a and 2a, respectively, are rigidly fixed by a massless, rigid rod of length $l = \sqrt{24}a$ through their centers. This assembly is laid on a firm and flat surface, and set rolling without slipping on the surface so that the angular speed about the axis of the rod is ω. The angular momentum of the entire assembly about the point 'O' is $\vec{L}$ (see the figure). Which of the following statement(s) is(are) true?

(A) The center of mass of the assembly rotates about the z-axis with an angular speed of $\omega/5$
(B) The magnitude of angular momentum of center of mass of the assembly about the point O is $81ma^2\omega$
(C) The magnitude of angular momentum of the assembly about its center of mass is $17ma^2\omega/2$*
(D) The magnitude of the Z component of L is $55ma^2\omega$

*: In precessing frame.



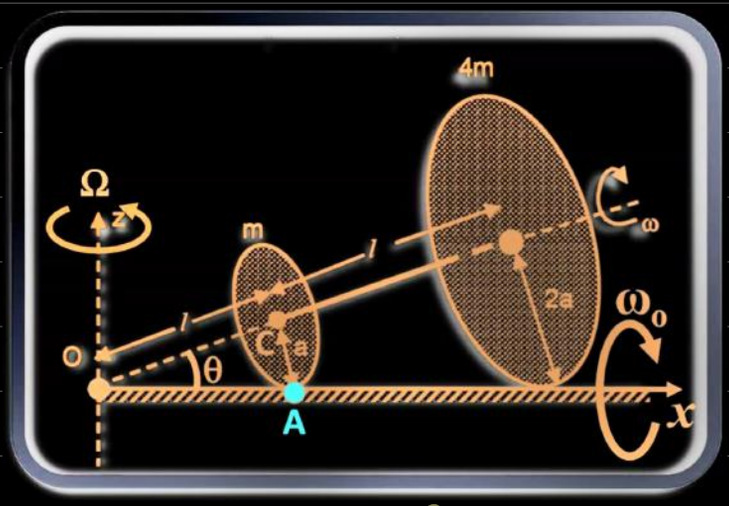


Clear Concepts With Ambarish

Solution: $l = \sqrt{24}\,a \Rightarrow OA = \sqrt{24a^2 + a^2}$

$\Rightarrow OA = 5a$. Now net angular velocity ω_0 is along instantaneous axis OA thus component of spinning angular velocity and precession angular velocity Ω must caniel $\perp$ to this direction.

Thus, $\Omega = \omega \sin\theta = \omega \times (1/5) = \frac{\omega}{5}$



Thus $\Omega = \omega/5$ (option a is correct).

Now distance of CM from O: $(m \times l + 4m \times 2l)/5m = \frac{9}{5}l$

Now speed of CM: $V_{CM} = \Omega \times \frac{9}{5} l \cos\theta = \frac{\omega}{5} \times \frac{9}{5} \times \sqrt{24}\,a \times \frac{\sqrt{24}\,a}{5a}$

$\Rightarrow V_{CM} = \frac{216}{125}\omega a$, Now angular momentum of CM about O: $\vec{r}_{CM} \times M\vec{V}_{CM}$

Thus required angular momentum of CM about O: $\frac{9}{5} l \times 5m \times \frac{216}{125}\omega a$

Thus, $|M(\vec{r}_{CM} \times \vec{V}_{CM})| = \frac{1944\sqrt{24}}{125} m a^2 \omega$ $\therefore$ option (B) is wrong.

Now In precessing frame, only spinning ω is visible thus angular momentum about CM in precessing frame is $I\omega$.

$I = \frac{1}{2}ma^2 + \frac{1}{2} \times 4m(2a)^2 = \frac{17}{2}ma^2 \Rightarrow L = \frac{17}{2}ma^2\omega$.

Thus option (C) is correct provided it means angular momentum in precessing frame.

Now to calculate angular momentum about z axis, we will first have to calculate angular momentum about point O considering principal axes and then we can take its component along z axis. This is shown in the following analysis:

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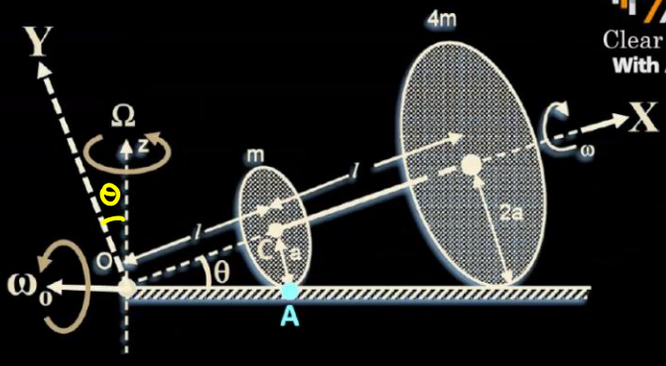
Angular Momentum about O:



$\omega_o = \omega\cos\theta = \frac{\sqrt{24}}{5}\omega \quad (\because \cos\theta : \frac{\sqrt{24}}{5})$

Now we introduce Axes X, Y & Z (see fig).

$\vec{\omega_o} = \omega_o(Cos[\theta](-\hat{I}) + Sin[\theta]\hat{J})$ ①

($\hat{I}$ and $\hat{J}$ are unit vectors along x and y).

Now:

$\vec{L_O} = I_X\omega_X\hat{\imath} + I_Y\omega_Y\hat{\jmath} + I_Z\omega_Z\hat{K}$

Thus, its vertical Component:

$L_{O_z} = I_X\omega_X Sin[\theta] + I_Y\omega_Y Cos[\theta]$ ②

$L_{O_z} = Sin[\theta]Cos[\theta]\omega_o(-I_X + I_Y)$ ($\because$ ① & ②)

$I_X = \frac{17}{2}ma^2, I_Y = \frac{1649}{4}ma^2$ ($\because$ ll axis theorem)

$L_{O_z} = \frac{\sqrt{24}}{25}ma^2\frac{\sqrt{24}}{5}\omega\left(\frac{17}{2} + \frac{1649}{4}\right)$

$L_{O_z} = \frac{1938}{25}a^2m\omega$ **Option D is incorrect**

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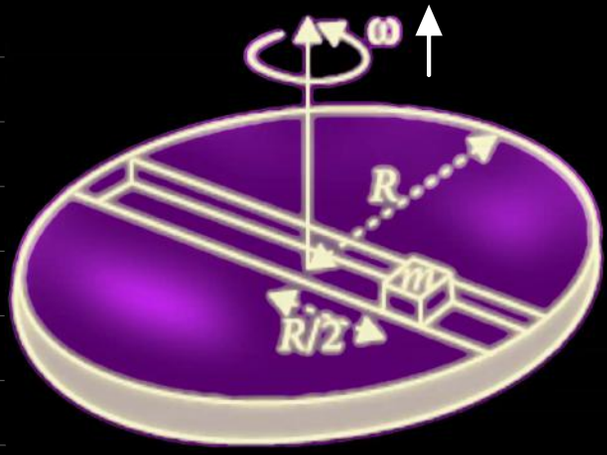
Example: JEE Advanced 2016:



A horizontal disc of radius R has a smooth rectangular slot cut in it along its diameter as shown. The disc is rotating with a constant angular velocity ω. A small block of mass m is placed gently in the slot at a distance R/2 from the centre. Find (a) distance of the block from the centre at a general time t (b) / Surface reactions on block as a function of time. Take radial direction along the groove as $\hat{n}$ and vertical as $\hat{k}$.

Solution: Here block moves due to centrifugal forces and experiences wall reaction due to coriolis force. Here from the rotating frame we can find distance as a function of time using concept of centrifugal potential Energy. Thus by conservation of mechanical energy:

$-\frac{1}{2}m\omega^2\left(\frac{R}{2}\right)^2 = -\frac{1}{2}m\omega^2 r^2 + \frac{1}{2}m v_{rot}^2$ where V_{rot} is the

speed as seen from rotating frame. Thus:

$$v_{rot} = \omega\sqrt{r^2-(R/2)^2} = dr/dt$$

$$\Rightarrow \int_{R/2}^{r} \frac{dr}{\sqrt{r^2-(R/2)^2}} = \int_0^t \omega\, dt \quad , \text{ Put } r = R/2 \sec\theta$$

$$\Rightarrow dr = \frac{R}{2}\sec\theta\tan\theta\, d\theta$$

$$\Rightarrow \int_0^{\sec^{-1}(2r/R)} \frac{R/2\ \sec\theta\tan\theta\, d\theta}{R/2\ \tan\theta} = \int_0^t \omega\, dt$$

$$\Rightarrow \int_0^{\sec^{-1}(2r/R)} \sec\theta\, d\theta = \omega t$$

$$\Rightarrow \left[\ln|\sec\theta+\tan\theta|\right]_0^{\sec^{-1}(2r/R)} = \omega t$$

$$\Rightarrow \frac{2r}{R}+\sqrt{\left(\frac{2r}{R}\right)^2-1} = e^{\omega t} \Rightarrow r = \frac{R}{4}\left[e^{\omega t}+e^{-\omega t}\right] \text{ (1) Ans.}$$

Now in rotating frame, coriolis force from wall: $\vec{F}_{cor} = -2m\,\vec{\omega}\times\vec{v}_{rot}$

Also $\vec{N}_{wall} + \vec{F}_{cor} = 0 \Rightarrow \vec{N}_{wall} = 2m\,\vec{\omega}\times\vec{v}_{rot} = 2m\omega^2\sqrt{r^2-(R/2)^2}\,\hat{\theta}$

Substituting r from (1): $\vec{N}_{wall} = \frac{1}{2}mR\omega^2\left(e^{\omega t}-e^{-\omega t}\right)\hat{\theta}$ and

$\vec{N}_{floor} = mg\,\hat{k}$ Ans.

→ Recall that net pseudo force in rotating frame has three terms: Coriolis: $-2m\vec{\omega}\times\vec{v}_{rot}$, Centrifugal: $-m\omega^2 r\hat{r} = -m\,\vec{\omega}\times(\vec{\omega}\times\vec{r})$ and Euler: $-m\vec{\alpha}\times\vec{r}$. Besides if centre of rotating frame has a translational acceleration $\vec{a}$, an additional pseudo force $-m\vec{a}$ will also act.

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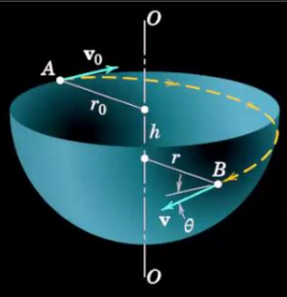
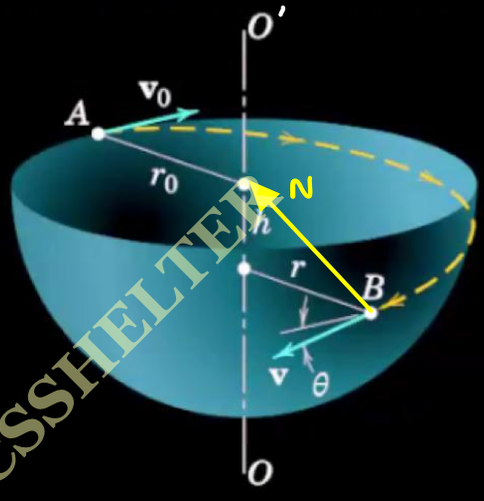
EXAMPLE : MERIAM KRAIGE :

A small mass particle is given an initial velocity $\mathbf{v}_0$ tangent to the horizontal rim of a smooth hemispherical bowl at a radius r_0 from the vertical centerline, as shown at point A. As the particle slides past point B, a distance h below A and a distance r from the vertical centerline, its velocity $\mathbf{v}$ makes an angle θ with the horizontal tangent to the bowl through B. Determine θ.





Solution: Here normal reaction always passes through the vertical axis OO' and mg is ll to this axis, thus there is no torque about vertical axis ∴ angular momentum about this axis must be conserved. Also mechanical energy must be conserved. Thus:

$$\frac{1}{2}mv_0^2 = \frac{1}{2}mv^2 - mgh \quad (1) \quad (\because \text{COME})$$

also: $mv_0 r_0 = mv\cos\theta\, r \quad (2)$

Solving (1) & (2), we get $\cos\theta = \dfrac{v_0 r_0}{\sqrt{(v_0^2 + 2gh)(r_0^2 - h^2)}}$ Ans.

→ Think! How will you find maximum depth reached by the particle?

Solution: At maximum depth, velocity will not have any vertical component thus it must be along horizontal tangent. Thus we can put $\theta = 0$ and solve for h to get maximum depth. (Infact the particle will oscillate between the rim and maximum depth periodically).

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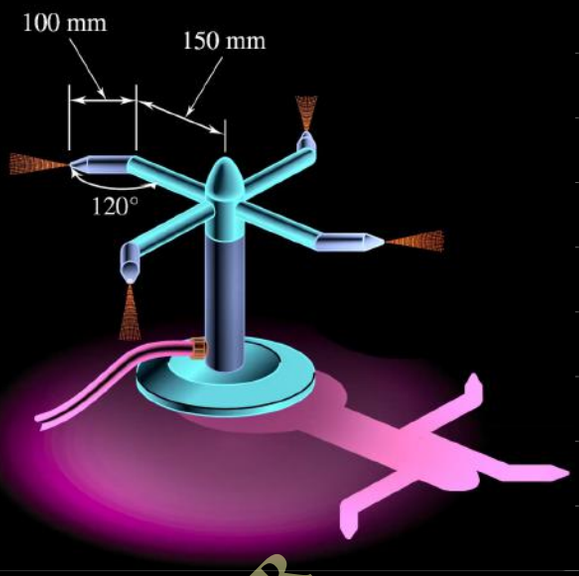
Angular Momentum Analysis Of Variable Particle Systems:



Just as in case of variable particle systems, net force on the liquid was equal to the rate of creation of linear momentum similarly, the net torque is equal to rate of creation of angular momentum. Further in steady mass flow case, in systems with single exit and entry, the torque on liquid is

$\vec{\tau} = \dot{m}\left[(\vec{r}_{out} \times \vec{v}_{out}) - (\vec{r}_{in} \times \vec{v}_{in})\right]$, In case of multiple inlets and outlets,

$\vec{\tau} = \sum \dot{m}_i (\vec{r}_i \times \vec{v}_i)$ where $\vec{r}_i$ is the position vector of i^{th} outlet, and $\vec{v}_i$ is velocity at i^{th} outlet.

This holds true when container is not having angular acceleration. In case container has angular acceleration, we also need to account for the angular momentum creation of the fluid inside the pipes due to angular acceleration of the container.

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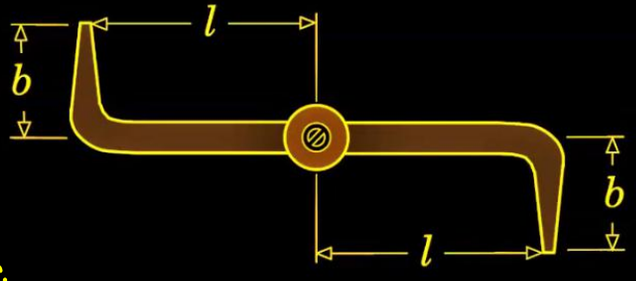
Example: Pathfinder: Check-41:

A water sprinkler system consists of a tube bent into S shape. This tube rotates in a horizontal plane about a vertical axis through its centre when water flows out from its nozzles. Top view of the tube is shown in the figure. Moment of inertia of the rotating tube including water about the axis is I, cross-section of outlet of a nozzle is S, density of water is ρ,
(a) Derive an expression for the initial angular acceleration of the tube.
(b) Derive an expression for the angular velocity ω of the tube as a function of time t.



Given exit speed of water is u relative to the nozzles.

Solution: Since there is no external torque about vertical axis, rate of creation of clockwise angular momentum due to ejection of water must be equal to rate of creation of anticlockwise angular momentum due to anti-clockwise angular acceleration of the sprinklers. Thus: initially: $(\rho S u^2 l \times 2) = I\alpha$ $(\because \dot{m}v = \rho S u^2)$.

$\Rightarrow \alpha = 2\rho S u^2 l / I$ Ans.

Now if ω is the anticlockwise angular velocity of the sprinkler, rate of creation of clockwise angular momentum (due to exiting water):

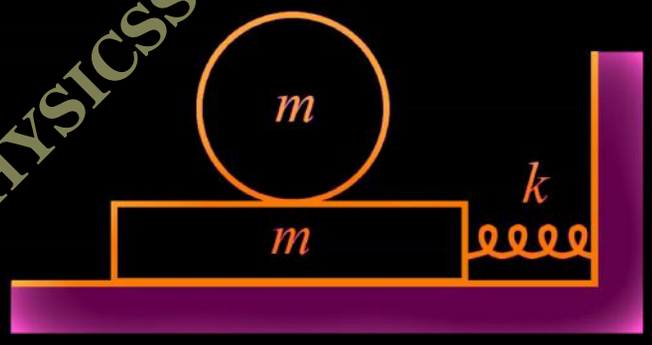
Thus:

$$I\frac{d\omega}{dt} = 2\rho s u(u-\omega l)\times l \qquad (\because \dot{m}=\rho s u,\ \perp \text{ velocity} = u-\omega l).$$

$$\Rightarrow \int_0^{\omega}\frac{d\omega}{u-\omega l} = \int_0^{t}\frac{2\rho s u l}{I}\,dt \Rightarrow \frac{-1}{l}\left[\ln(u-\omega l)\right]_0^{\omega} = \frac{\rho s u l t}{I}$$

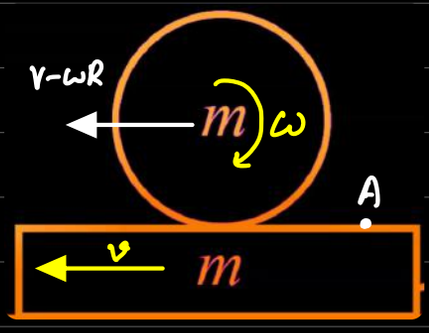
$$\Rightarrow \ln\left(\frac{u-\omega l}{u}\right) = \frac{-2\rho s u l^2 t}{I} \Rightarrow \omega = \frac{u}{l}\left(1-e^{-2us\rho l^2 t/I}\right) \quad \text{Ans.}$$

EXAMPLE: On a smooth surface, a plank compresses a spring of constant k through a distance A. A rough cylinder of same mass as plank is placed on the plank and the system is released from rest. What is the velocity of the plank when the spring attains its natural length assuming that the cylinder does not slip.



Solution: Let speed of plank be v and angular speed of cylinder be ω. The angular momentum of cylinder remains conserved about any axis on plank perpendicular to plane of motion. Thus:

CAM: $\frac{1}{2}mR^2\omega - m(v-\omega R)R = 0$



$$\Rightarrow \omega = 2v/3R \Rightarrow v_{cm} = v-\omega R = v/3.$$

Now COME: $\frac{1}{2}kA^2 = \frac{1}{2}mv^2 + \frac{1}{2}m\left(\frac{v}{3}\right)^2 + \frac{1}{2}\cdot\frac{1}{2}mR^2\left(\frac{2v}{3R}\right)^2$

$$\Rightarrow v = A\sqrt{\frac{3k}{4m}} \quad \text{Ans.}$$

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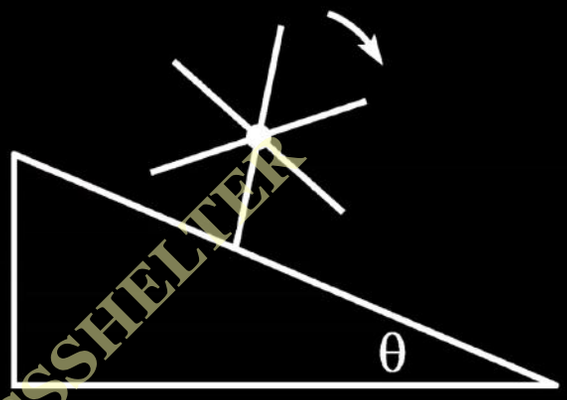
→ Once you learn SHM, you will know that $v = A \times 2\pi/T$ where T is time period of oscillations. Thus $T = 2\pi\sqrt{4m/3k}$.

Example: David Morin:

Pencil on a plane ****

This exercise deals with the terminal velocity of a "pencil" rolling down a plane. To simplify things, we'll assume that the pencil has all its mass on its axis. And to avoid messy complications, we'll assume that the cross section of the pencil looks like a wheel with six equally spaced massless spokes and no rim (see Figure). Let the length of the spokes be r, and let the plane be inclined at an angle θ. Assume that there is sufficient friction to prevent the spokes from slipping on the plane, and assume that the pencil does not bounce when a spoke hits the plane.



(a) Explain qualitatively why the pencil reaches a terminal (average) velocity, assuming that it remains in contact with the plane at all times.

(b) Assume that conditions have been set up so that the pencil eventually reaches a nonzero terminal (average) velocity, while remaining in contact with the plane at all times. Describe this terminal velocity. You may do this by stating the maximum speed of the axis in the limiting steady state.

(c) What is the minimum value of θ for which a nonzero terminal velocity exists? An initial kick to the pencil is allowed.

(d) What is the maximum value of θ for which the pencil remains in contact with the plane at all times? As a check on your answer, the difference between the answers in parts (c) and (d) is about 5.09°.

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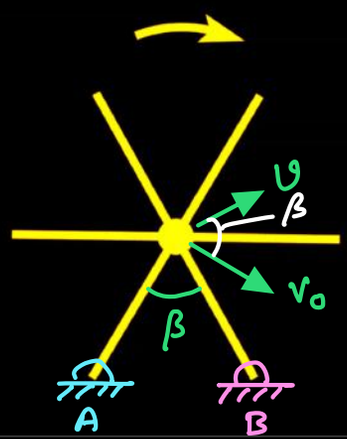
Solution:

(a) The main point is that when the pivot point of the pencil changes (when a new spoke hits the plane), the speed of the axis changes suddenly and kinetic energy is lost, because only the velocity component perpendicular to the new spoke survives from the previous velocity, which was perpendicular to the old spoke. (Equivalently, in the collision where a new spoke hits the plane, angular momentum is conserved around the point of impact.) The loss in kinetic energy is proportional to the square of the velocity right before the change of spoke. When the speed has increased to a magnitude where this loss in kinetic energy equals the gain from the change in potential energy, the pencil will not go any faster.



See Figure: A: Pre collision Pivot
B: Post collision Pivot.

Due to collision at B angular momentum about B is conserved.
Thus component of velocity ⊥ to B is same as pre collision component.
Thus $v = v_0 \cos\beta$. v_0: Pre collision speed, v: post collision speed.

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(b) Let's solve this problem for a general number of spokes, N, and then let $N = 6$. Let v_0 be the speed of the axis right before a new spoke hits, and let $\beta \equiv 2\pi/N$. Then the speed of the axis right after the new spoke hits is $v_0 \cos\beta$, because this is the component of the old velocity that is perpendicular to the new spoke. The length of a "side" of the pencil is $2r\sin(\beta/2)$, so equating the change in potential energy during an Nth of a rotation with the kinetic energy loss due to the changing of the contact spoke (because these two quantities will balance in the "steady" state) gives

$$(mv_0^2/2)(1 - \cos^2\beta) = mg\big(2r\sin(\beta/2)\big)\sin\theta.$$

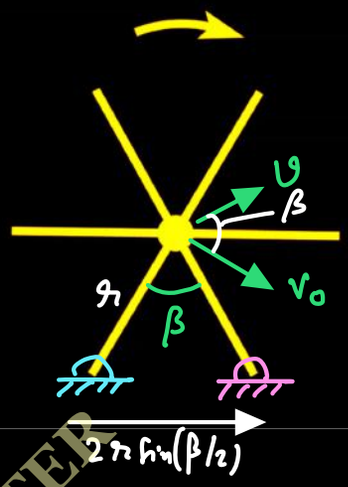
Therefore, in the steady state the maximum speed v_0 of the axis is given by

$$v_0^2 = \frac{4gr\sin(\beta/2)\sin\theta}{\sin^2\beta}. \quad (1)$$

For $N = 6$ and $\beta = \pi/3$, this yields

$$v_0^2 = (8/3)gr\sin\theta.$$

If conditions have been set up so that a nonzero v_0 exists, then it must be this (assuming that contact is always maintained with the plane).



(c) If $\theta < \beta/2$, then right after the pivot point changes, the axis must actually move upward before falling down along the plane. For a nonzero v_0 to exist, the axis must be moving fast enough to get over this "bump" (remember that an initial kick to the pencil is allowed). Assuming $\theta < \beta/2$, the height that the axis must climb is $r\big(1 - \cos(\beta/2 - \theta)\big)$. The speed at which the axis starts this climb is $v_0\cos\beta$. Therefore, we must have $(1/2)m(v_0\cos\beta)^2 > mgr\big(1 - \cos(\beta/2 - \theta)\big)$. Using the expression for v_0 in Eq. (1), this gives

$$\frac{2\sin(\beta/2)\sin\theta\cos^2\beta}{\sin^2\beta} > 1 - \cos(\beta/2 - \theta).$$

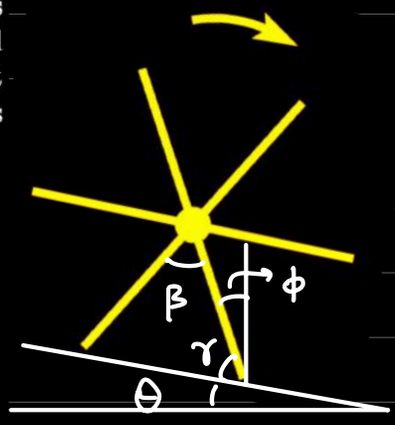
For $N = 6$ and $\beta = \pi/3$, this becomes

$$\frac{\sqrt{3}}{2}\cos\theta > 1 - \frac{5}{6}\sin\theta.$$

Squaring and solving for $\sin\theta$ gives

$$\sin\theta > \frac{15 - 6\sqrt{3}}{26}.$$

(The other root doesn't satisfy $\theta < \beta/2 = 30°$.) This minimum θ is approximately $10.2°$.



$\theta + \gamma + \phi = 90°$

$\Rightarrow \phi = 90° - \theta - (90° - \beta/2)$

$\Rightarrow \phi = \beta/2 - \theta$

(If ϕ is negative then CM does not have to climb over the bump).

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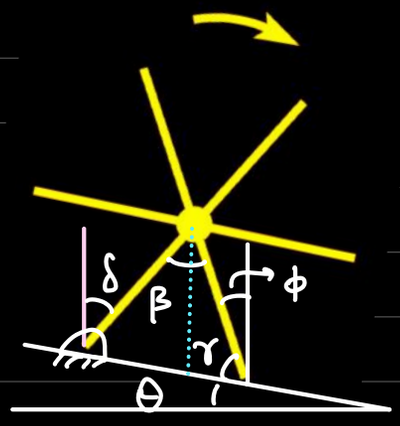
(d) The axis of the pencil moves in a circular arc around the pivot point. The gravitational force along the contact spoke must account (at least) for the centripetal acceleration of the axis. The maximal centripetal acceleration occurs right before the pivot point changes, and it equals mv_0^2/r. The minimal gravitational force along the spoke also occurs right before the pivot point changes, and it equals $mg\cos(\theta+\beta/2)$. Using the expression for v_0 in Eq. (①), the requirement $mv_0^2/r \le mg\cos(\theta+\beta/2)$ becomes

$$\frac{4\sin(\beta/2)\sin\theta}{\sin^2\beta} \le \cos(\theta+\beta/2) \implies \tan\theta \le \frac{\sin^2\beta}{4+\sin^2\beta}\cot(\beta/2).$$

For $N=6$ and $\beta=\pi/3$, this gives

$$\tan\theta \le \frac{3\sqrt{3}}{19},$$

which yields $\theta \lesssim 15.3°$. There is therefore a window of only about 5.1° (or 5.09°, to be a little more exact) for which the pencil has a nonzero terminal (average) velocity while remaining in contact with the plane at all times.



$\delta + \phi = \beta$

$\Rightarrow \delta = \beta - \phi$

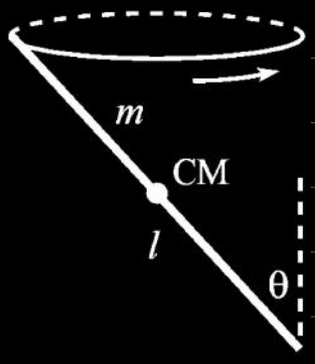
$= \beta - (\beta/2 - \theta)$

$\Rightarrow \delta = \beta/2 + \theta$

EXAMPLE: DAVID MORIN:

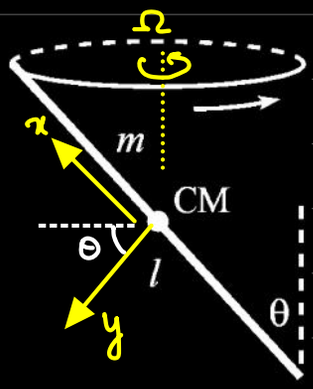
Stick under a ring **

A stick of mass m and length ℓ is arranged to have its CM motionless while its top end slides in a circle on a frictionless ring, as shown in Figure. The stick makes an angle θ with the vertical. What is the frequency of this motion?



Solution: Since CM does not accelerate, the ring must not offer any horizontal reaction and vertical reaction must be equal to mg. (Rod must be sliding using some hook type of arrangement). Now angular momentum about CM:

$\vec{L} = I_x\omega_x\hat{i} + I_y\omega_y\hat{j} + I_z\omega_z\hat{k}$ i.e



$$\vec{L} = -\frac{ml^2}{12}\Omega\sin\theta\,\hat{j} \qquad (\because I_x = 0 \;\&\; \omega_z = 0)$$

Now horizontal component of $\vec{L}$ precesses with Ω thus rate of change of angular momentum about O:

$$\frac{dL}{dt} = \frac{ml^2}{12}\Omega\sin\theta\cos\theta \times \Omega = \frac{ml^2}{12}\Omega^2\sin\theta\cos\theta.$$

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This must be equal to the torque of normal reaction about CM.

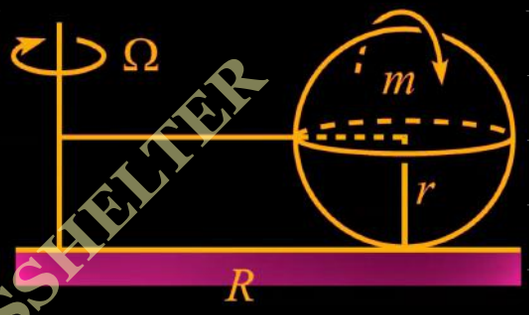
i.e $mg\frac{l}{2}\sin\theta = \frac{ml^2}{12}\Omega^2 \sin\theta\cos\theta \Rightarrow \Omega = \sqrt{\frac{6g}{l\cos\theta}}$

$\Rightarrow \nu = \Omega/2\pi = \frac{1}{2\pi}\sqrt{\frac{6g}{l\cos\theta}}$ Ans.

EXAMPLE: DAVID MORIN:

Rolling lollipop *

Consider a lollipop made of a solid sphere of mass m and radius r that is radially pierced by a massless horizontal stick. The free end of the stick is pivoted on a pole (see Figure), and the sphere rolls on the ground without slipping, with its center moving in a circle of radius R with frequency Ω. What is the normal force between the ground and the sphere?



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Solution: Here it is not given whether the ground is smooth or rough, however we can argue that even if ground were rough, there will be no need for friction for such a motion. If the surface is smooth, centre of mass will have only constant centripetal acceleration due to force from the stick which is $\perp$ to velocity of CM. Also due to conservation of mechanical energy, spinning angular velocity must remain constant. Thus roughness of floor will not matter here. Now $\omega_s r = \Omega R$ ($\because$ velocity of contact with ground is zero). Now horizontal component of angular momentum precesses due to torque of weight and normal reaction about the pivot. Thus:

$$I\omega_s\Omega = (N-mg)R \Rightarrow N = \frac{I\omega_s\Omega}{R} + mg = \frac{2}{5}mr^2 \times \frac{\Omega R}{r} \times \frac{\Omega}{R} + mg$$

Thus: $N = mg + \frac{2}{5}mr^2\Omega^2$ Ans.

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